



Alexandria University
Faculty of Veterinary Medicine
Anatomy & Embryology Department

جامعة الإسكندرية
ALEXANDRIA
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Faculty Of Veterinary Medicine كلية الطب البيطري



Mohamed Alsafy

Anatomy of the respiratory system

Prof. Mohamed Alsafy



larynx

It is short cartilaginous tube that connected the nasopharynx with trachea.

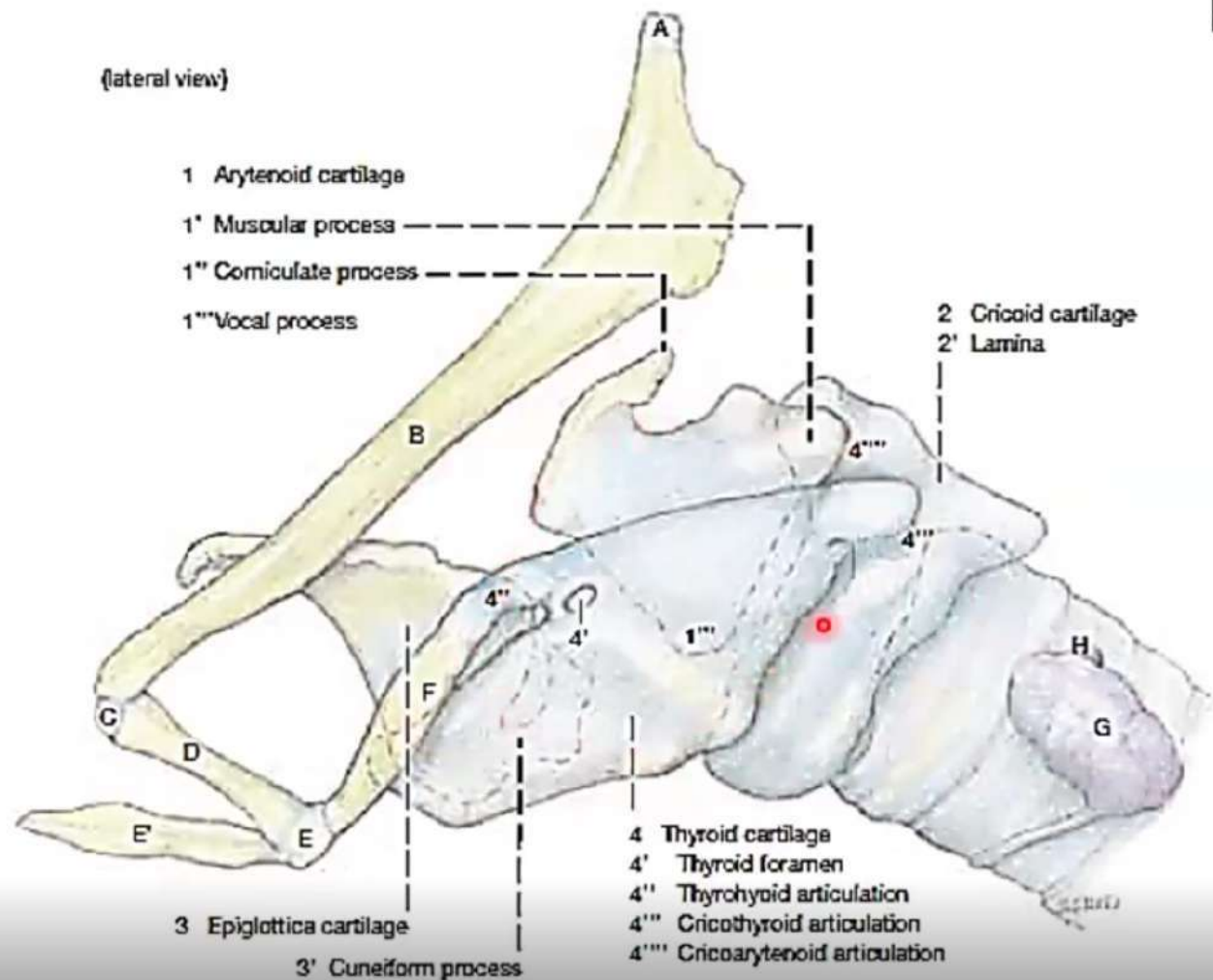
Position

At the level of the base of cranium, ventral to beginning of esophagus, within intermandibular space.

Larynx

(lateral view)

- A Tympanohyoid cartilage
- B Stylohyoid bone
- C Epithyoid bone
- D Ceratohyoid bone
- E Basihyoid bone
- E' Lingual process
- F Thyrohyoid bone
- G Thyroid gland
- H Parathyroid gland



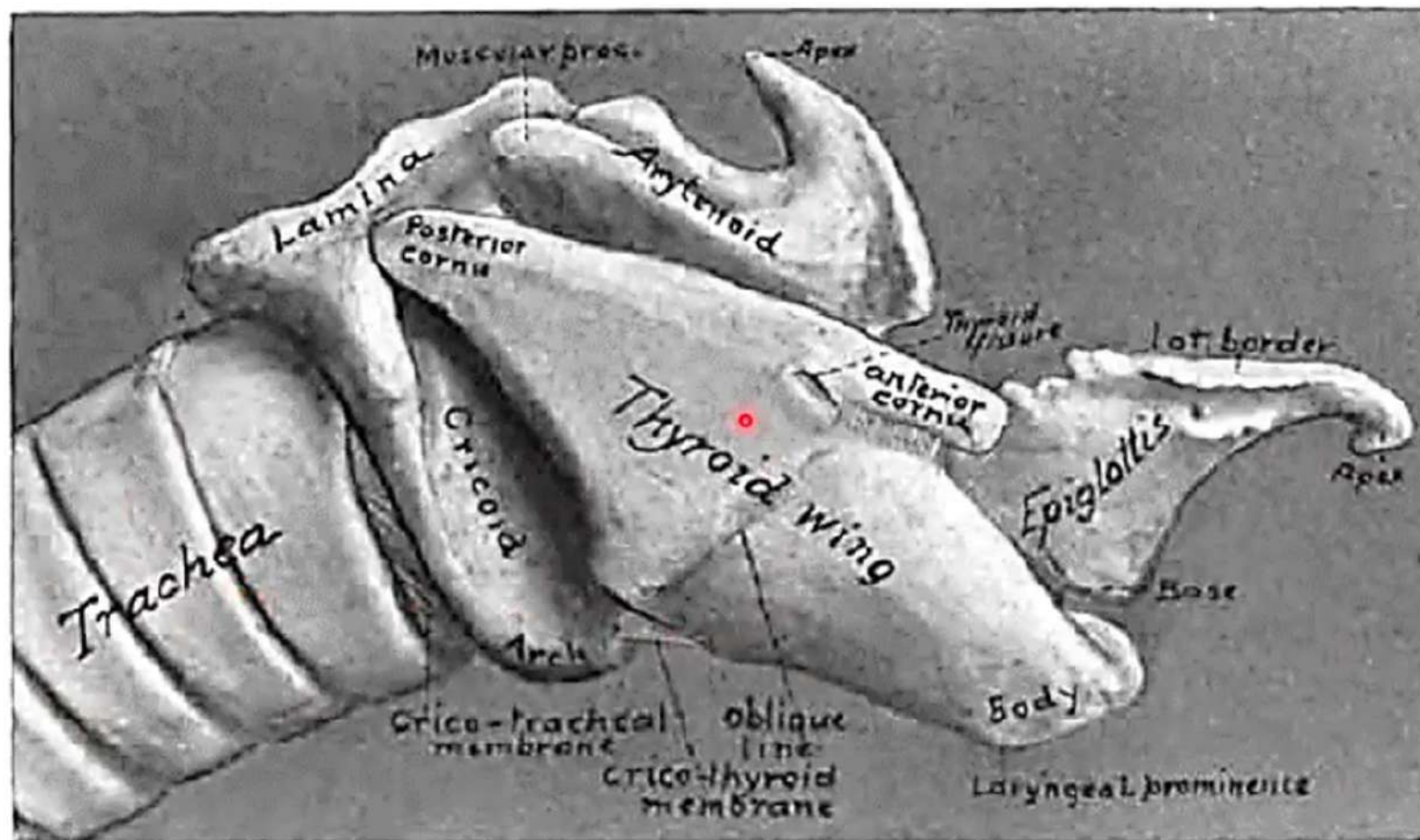


FIG. 349.—LARYNGEAL CARTILAGES OF HORSE, RIGHT VIEW.

2:28



Classification according to number

A- Single (thyroid, cricoid, epiglottis)

B-Paired (arytenoid, corniculate, cuneiform)

Classification according to structure

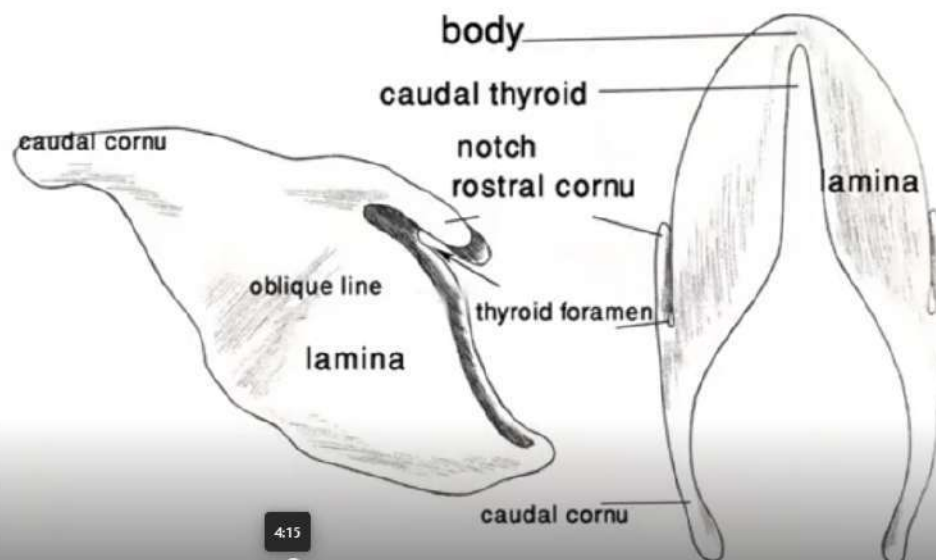
A- Hyaline (thyroid, cricoid, arytenoid)

B-Elastic (epiglottis, corniculate, cuneiform)

2:59

1. Thyroid cartilage

- Single and hyaline
 - a. body (ventrally) contain Adams apple.
 - b. two lamina (laterally).



4:15

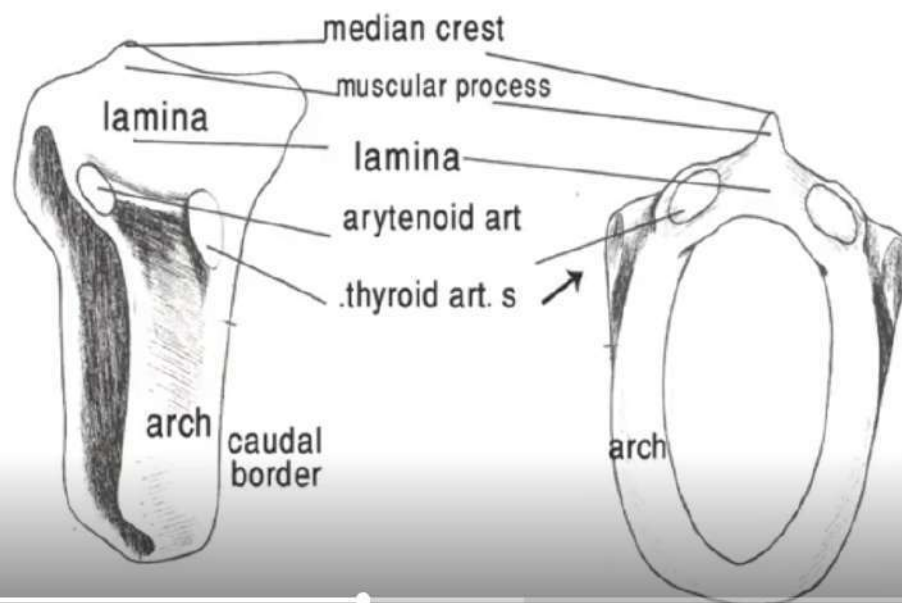
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2. Cricoid cartilage

- Single and hyaline
- Like signet ring.

a. lamina: dorsally.

b. arch

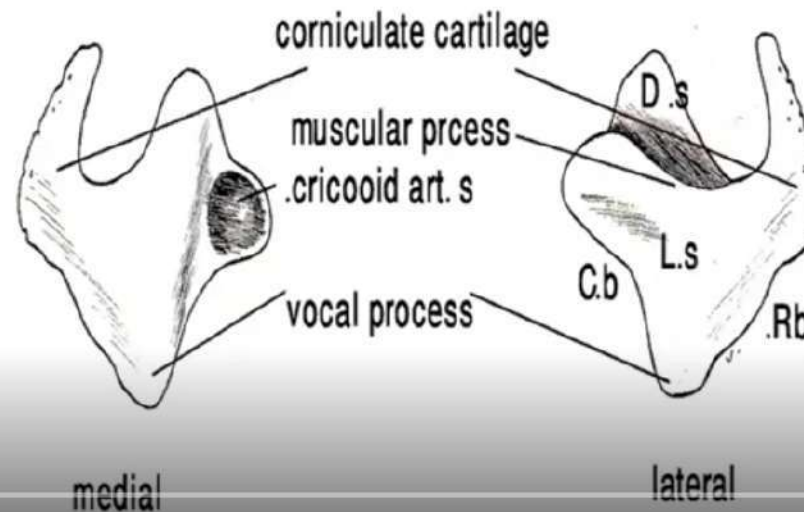




3. Arytenoid cartilage

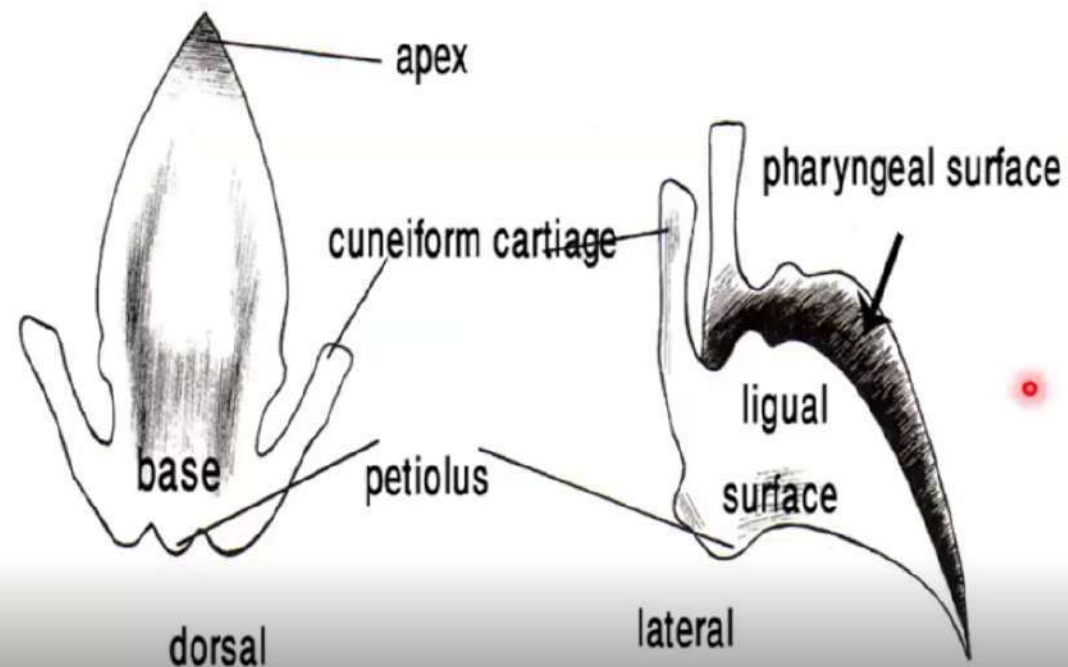
- Paired and hyaline.
- present dorsal to thyroid.
- pyramidal in shape, vocal process.

4. Corniculate cartilages



5. Epiglottis cartilage

- Single and fibroelastic,
- Rostral cartilage of larynx.
- Like leaf



Muscles of the larynx

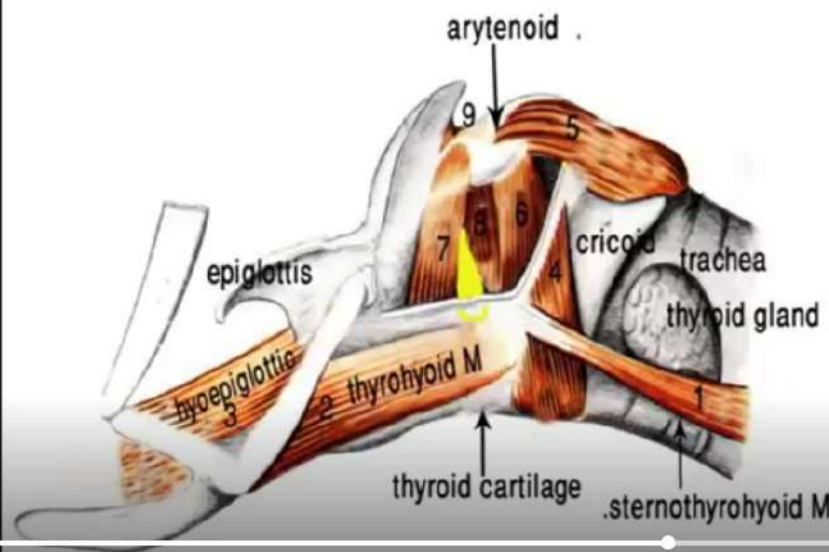
The laryngeal muscles are divided into

I. Extrinsic M.

1. Sternothyroid.
2. Thyrohyoid.
3. Hyoepiglottic.

II. Intrinsic M.

1. Cricothyroid.
2. Cricoarytenoideus.
 - a. Dorsalis
 - b. Lateralis.
3. Transverse arytenoid.
4. Thyroarytenoideus
- M. vesticularis.
- M.vocalis.

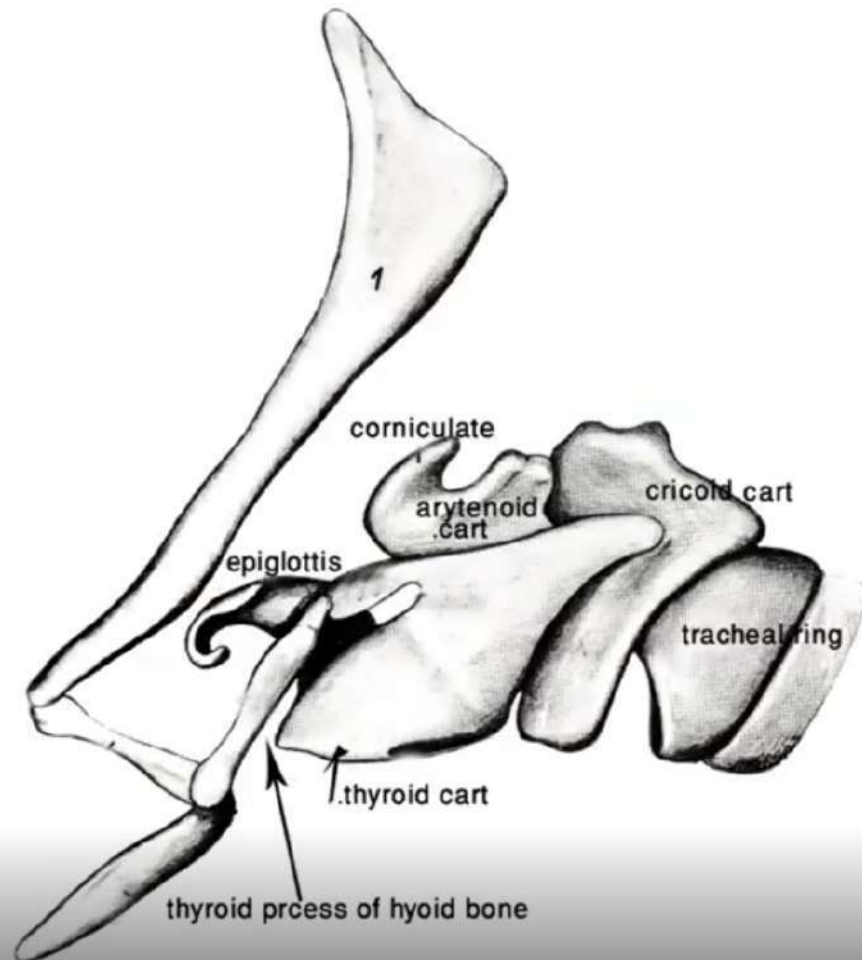


Laryngeal articulation

1.Cricothyroid joint

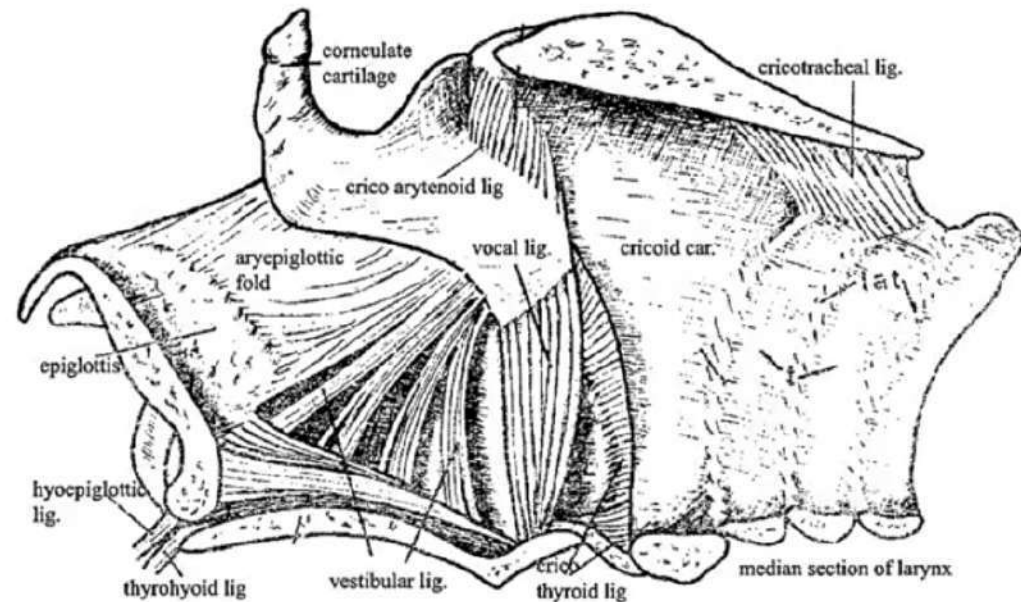
2.Cricoarytenoid joint

3.Thyrohyoid joint



Ligament of larynx

1. Cricotracheal.
2. Cricoarytenoid.
3. Cricothyroid.
4. Thyroepiglottic.
5. Thyrohyoid.
6. Hyoepiglottic.
7. Vestibular.
8. Vocal.
9. Transverse arytenoid.



Mucous membrane of the larynx



1. Aryepiglottic fold

2. Vocal fold

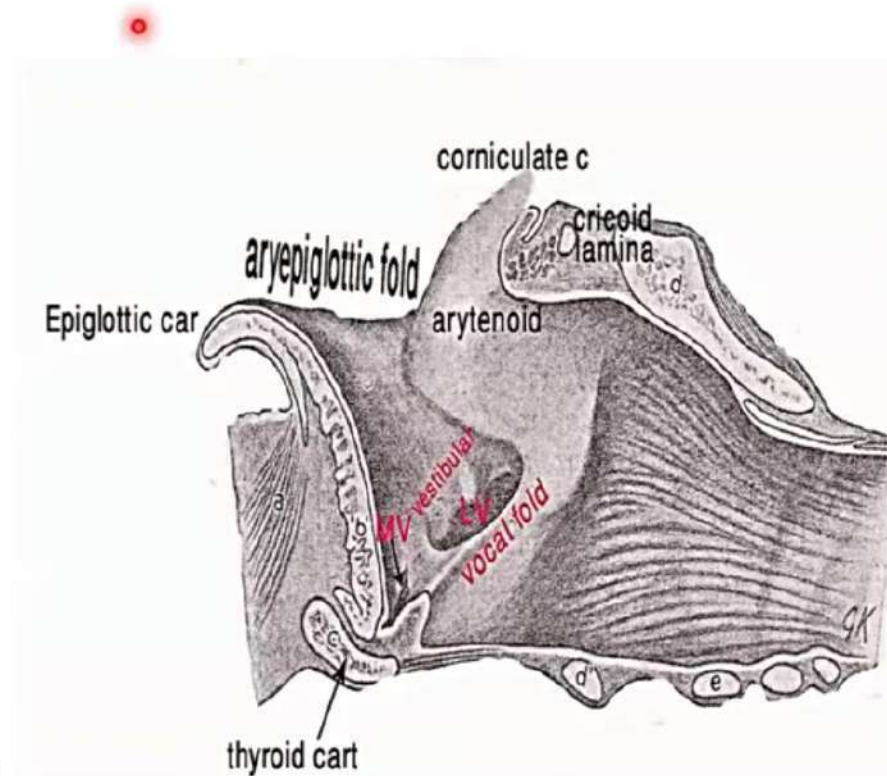
3. Vestibular fold

4. Lateral ventricle and laryngeal saccule

Depression of bet. Vocal & vestibular folds lead to Pouch called laryngeal saccule

5. Median ventricle

Depression at the base of epiglottis or floor of vestibule



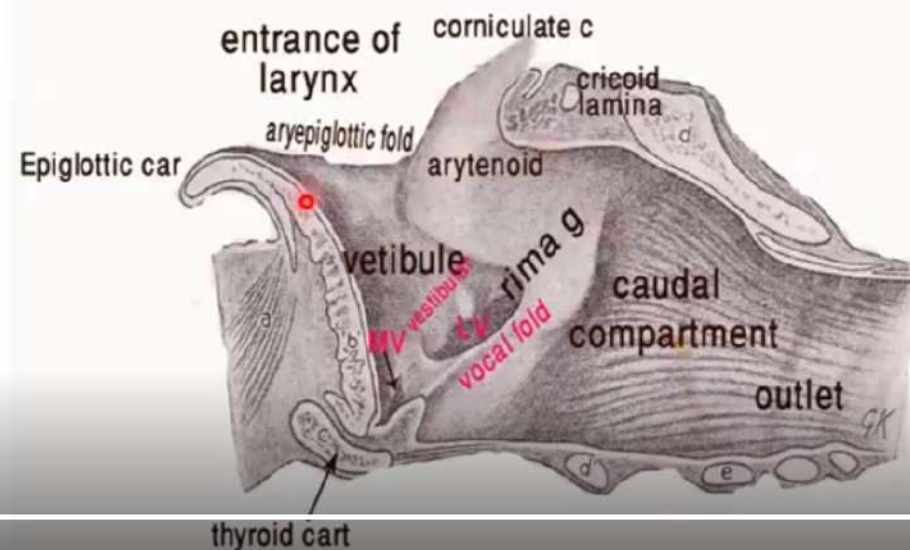


- ❖ **true vocal cord**= Vocalis muscles + vocal ligament + mucous membrane vocal cord.
- ❖ **false vocal cord**= Vestibular muscle + vestibular ligament + mucous membrane vestibular fold.

12:48

Cavity of the larynx

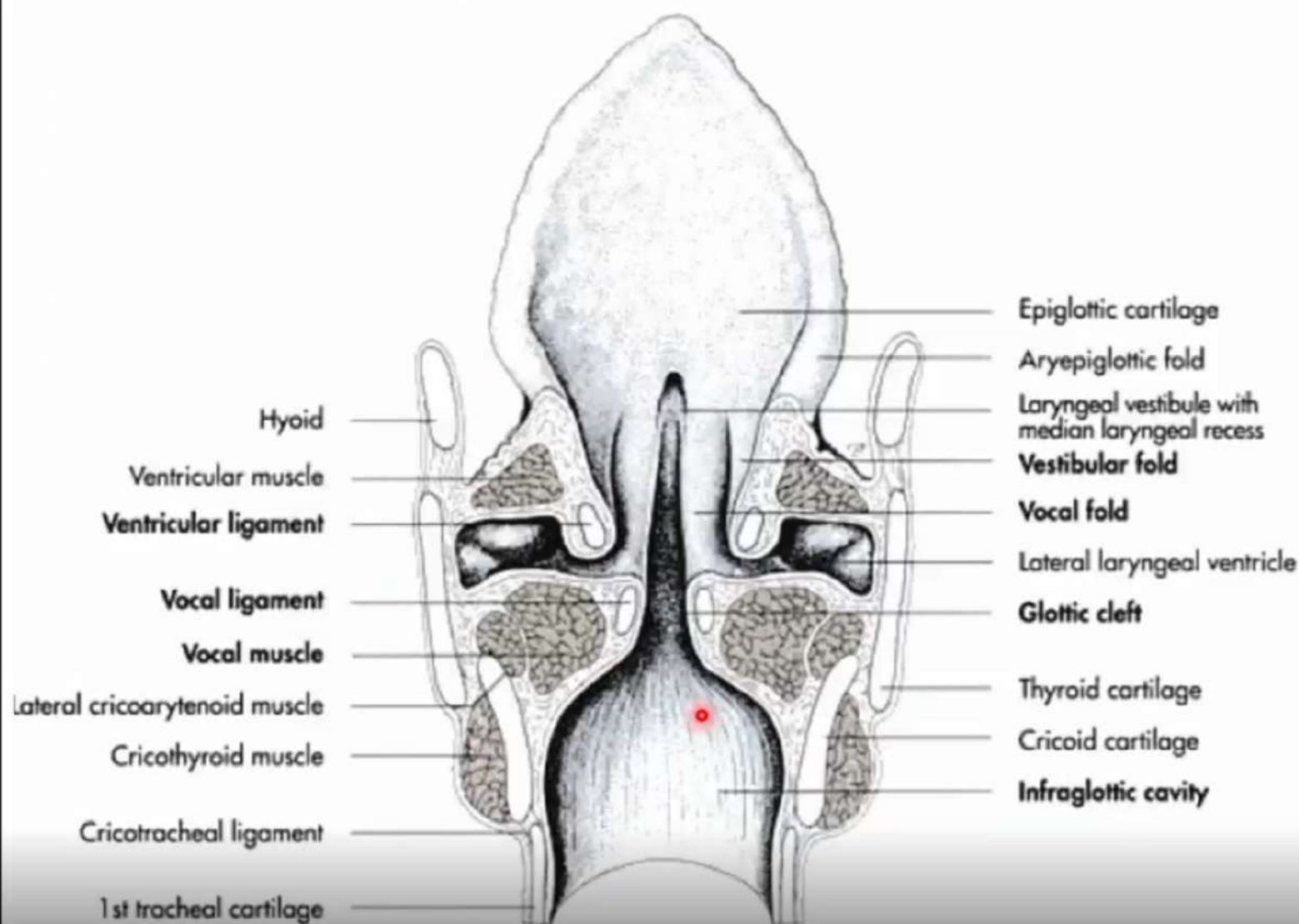
1. Entrance of larynx
2. Vestibule of larynx
3. Rima glottis
4. Caudal compartment of larynx
5. Outlet of larynx



13:49



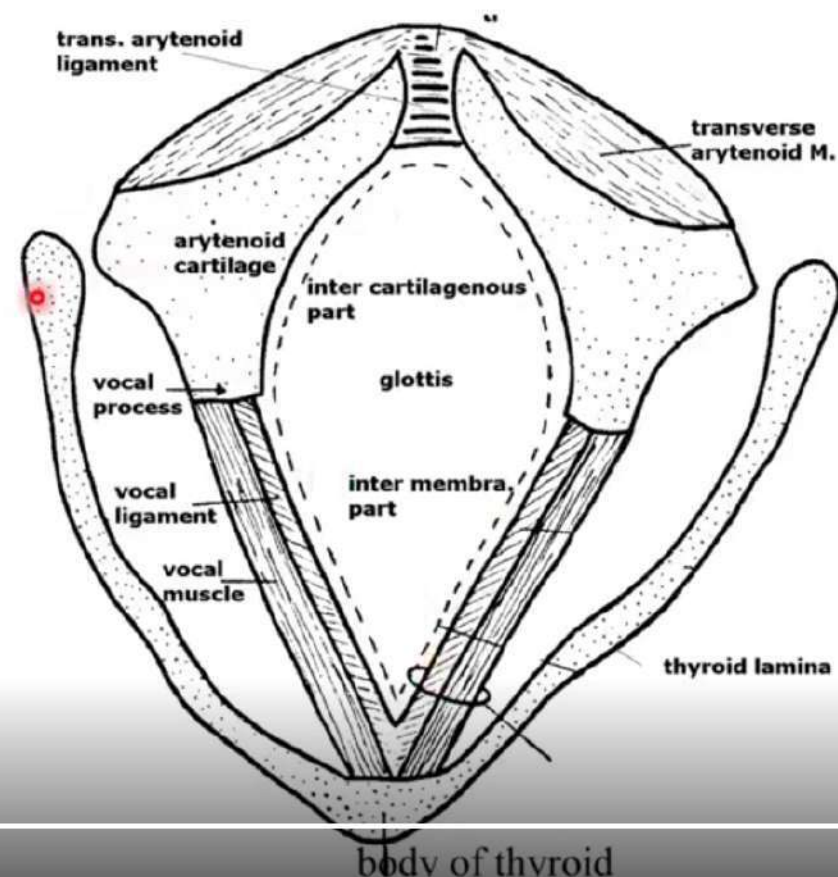
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14:39



- Rima glottis is the cleft bet. Vocal folds
- narrowest part of larynx
- Dorsal part intercartilaginous
- Ventral part intramembranous
- Glottis = vf + v.p. of ar. + r.g



14:54

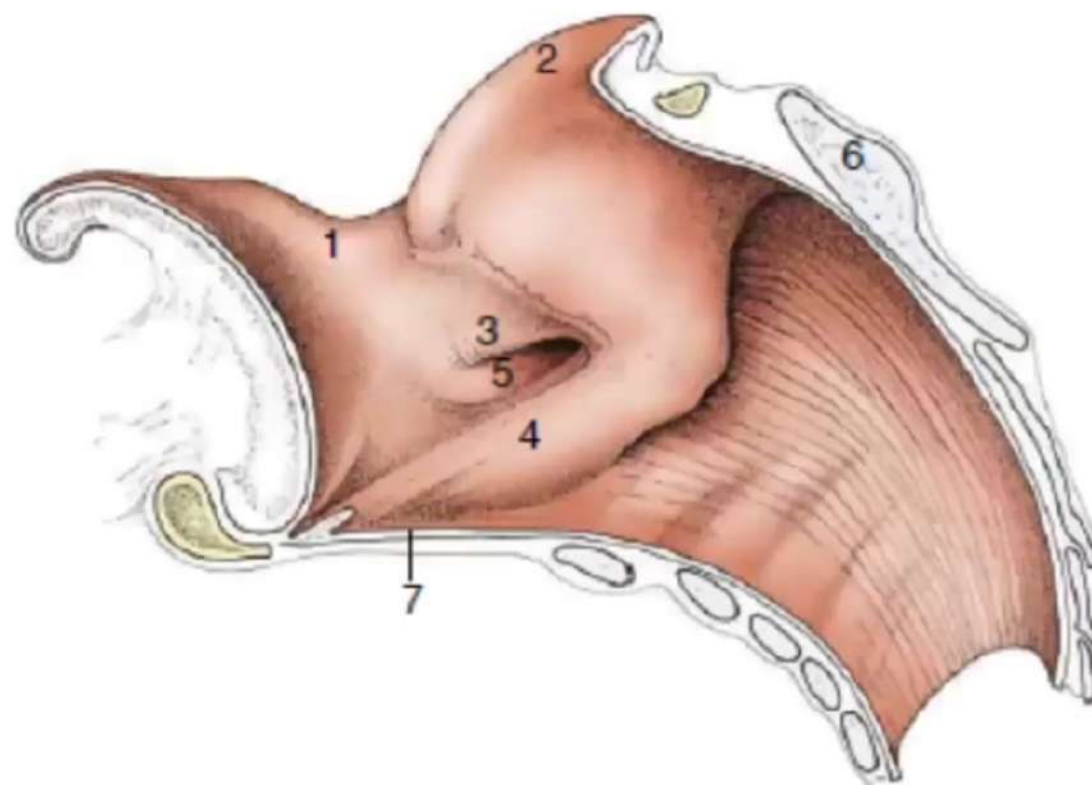
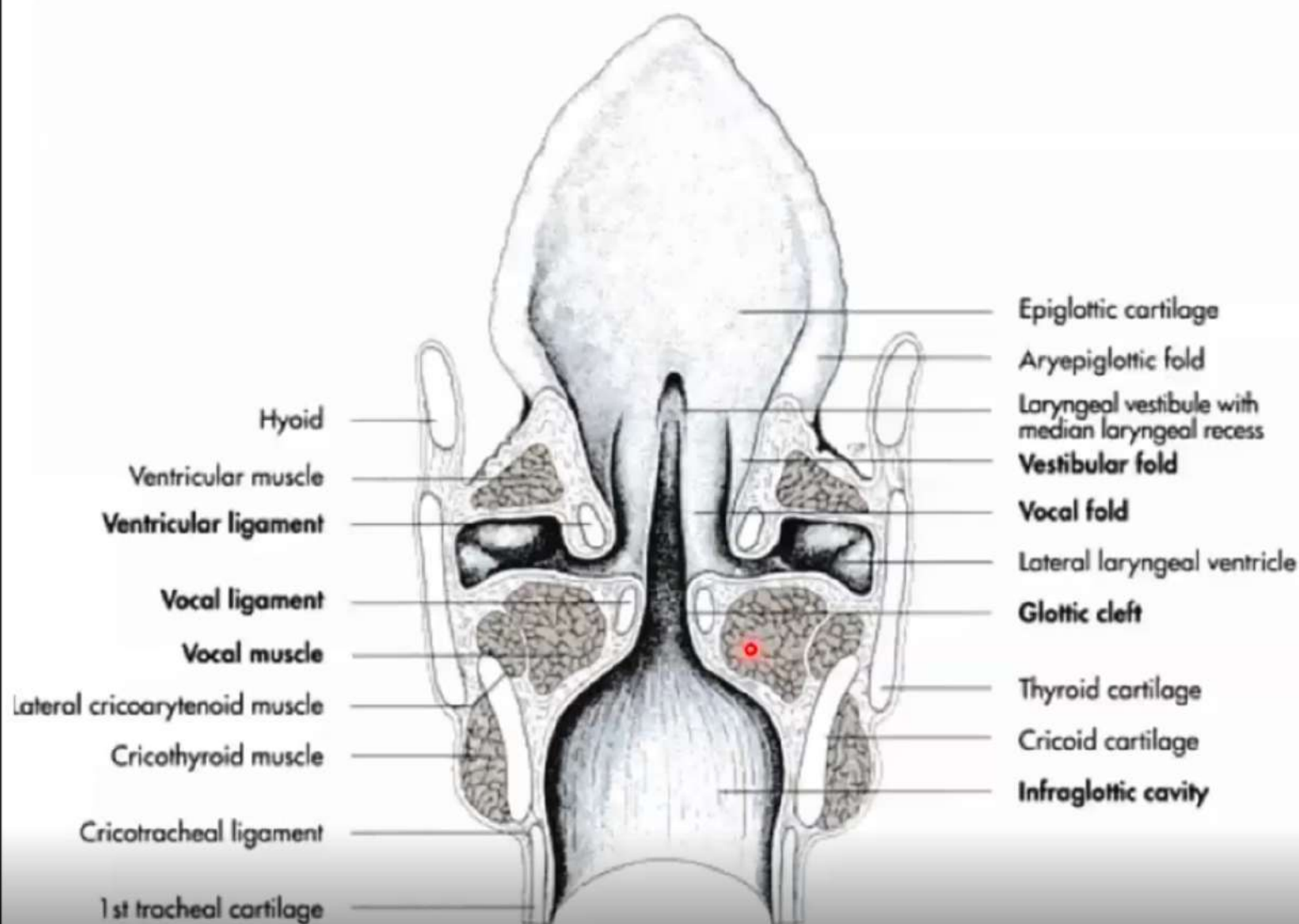


Figure 4-13 Median section of the equine larynx. 1, Epiglottis; 2, corniculate process of arytenoid cartilage; 3, vestibular fold; 4, vocal fold; 5, laryngeal ventricle; 6, lamina of cricoid cartilage; 7, cricothyroid ligament.

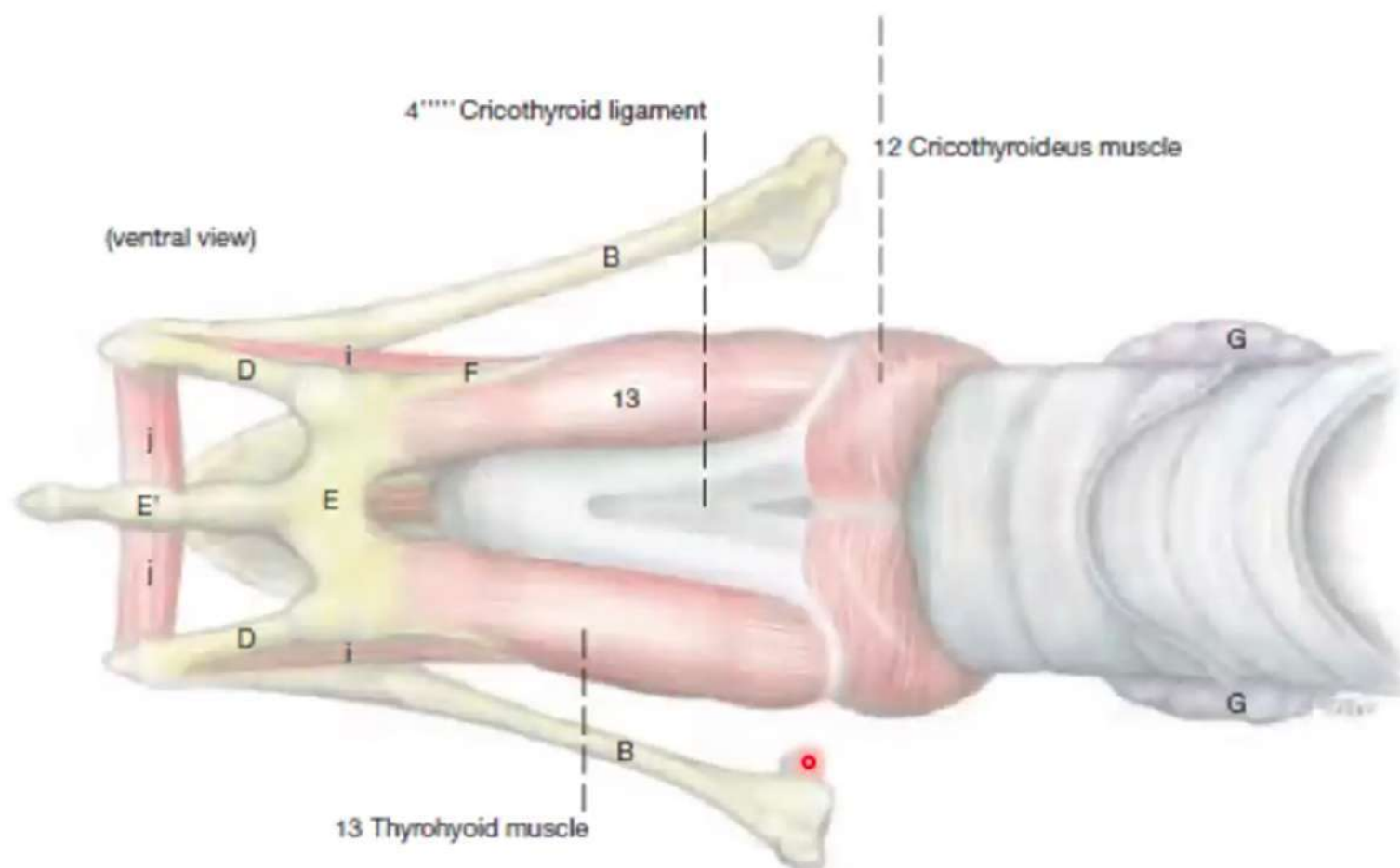


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- Cranial laryngeal nerve
- a Internal branch
- b External branch
- c Common carotid a.
- d Caudal thyroid a.
- e Cranial thyroid a.
- f Laryngeal branch
- g Ascending pharyngeal a.
- h Cranial laryngeal a.
- i Cricothyroid m.
- j transverse thyrod m.



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Special and Comparative Anatomy

Prepared by

Prof. Dr. Mohamed Alsafy

Head of Anatomy & Embryology Department

Anatomy of the respiratory system

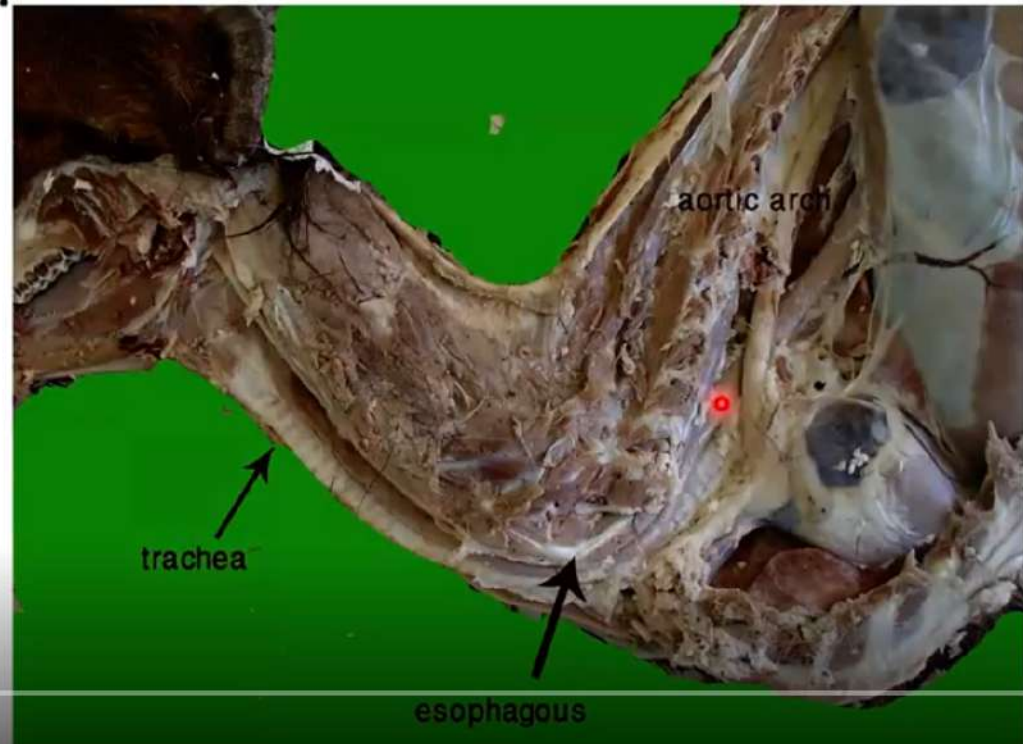
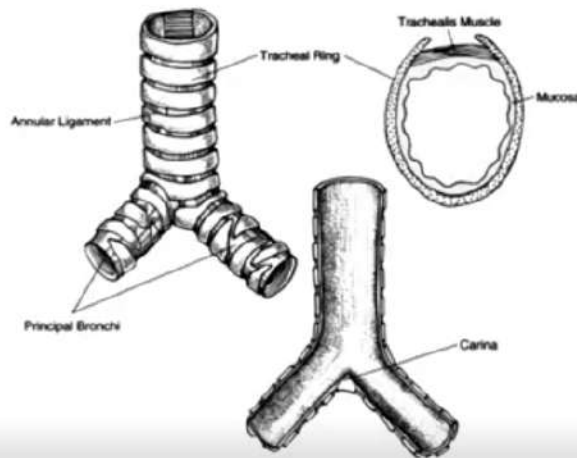
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Trachea



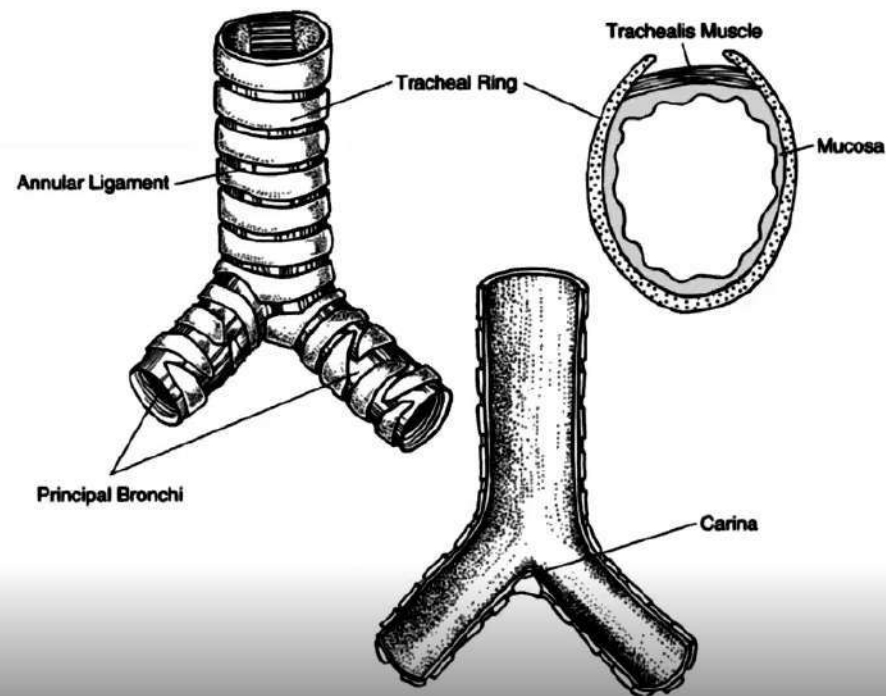
- It is a cartilaginous non-collapsible tube continued caudally from the cricoid cartilage of larynx to the root of lungs where it bifurcates to right and left principles bronchus.



Trachea (trachea)

The trachea extends from the cricoid cartilage of the larynx to its bifurcation. It consists of a series of C-shaped hyaline cartilages connected by ligaments. The number of the tracheal cartilages varies among individuals:

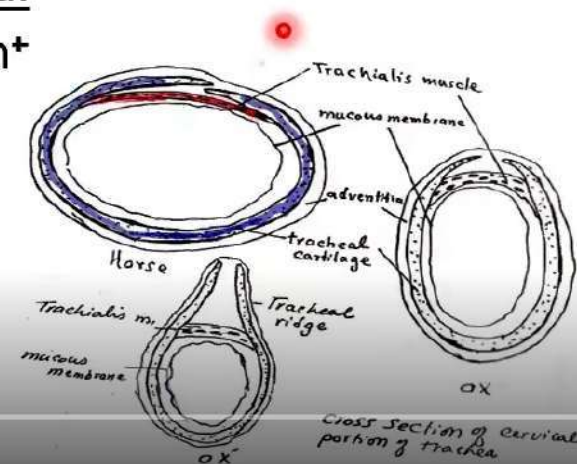
- horse: 48 – 60
- ox 48 – 60
- sheep: 48 – 60
- goat: 48 – 60
- dog: 42 – 46
- cat: 38 – 43





Comparative of Trachea

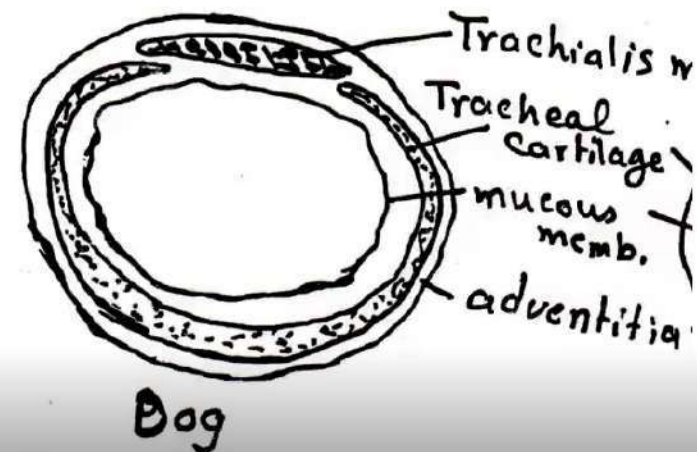
Horse	Ox	Camel
• Cylindrical • 48-60 T-rings • free ends of T-rings <u>overlap</u> • tracheal muscle is attached <u>inside</u> the free ends of the cartilaginous rings	• Oval • 48-60 T-rings • Free ends of T-rings meet dorsally forming <u>tracheal ridge</u> • tracheal muscle is attached <u>inside</u> the free ends of the cartilaginous rings • trachea gives <u>tracheal bronchus</u> to right apical lobe	• Circular • 48-60 T-rings • free ends of T-rings <u>overlap</u> • fusion between adjacent two or three rings • <u>tracheal bronchus</u>



Dog



- Cylindrical
- 42-46 T-rings
- free ends of T-rings widely separated dorsally
- tracheal muscle is attached outside the free ends of the cartilaginous rings



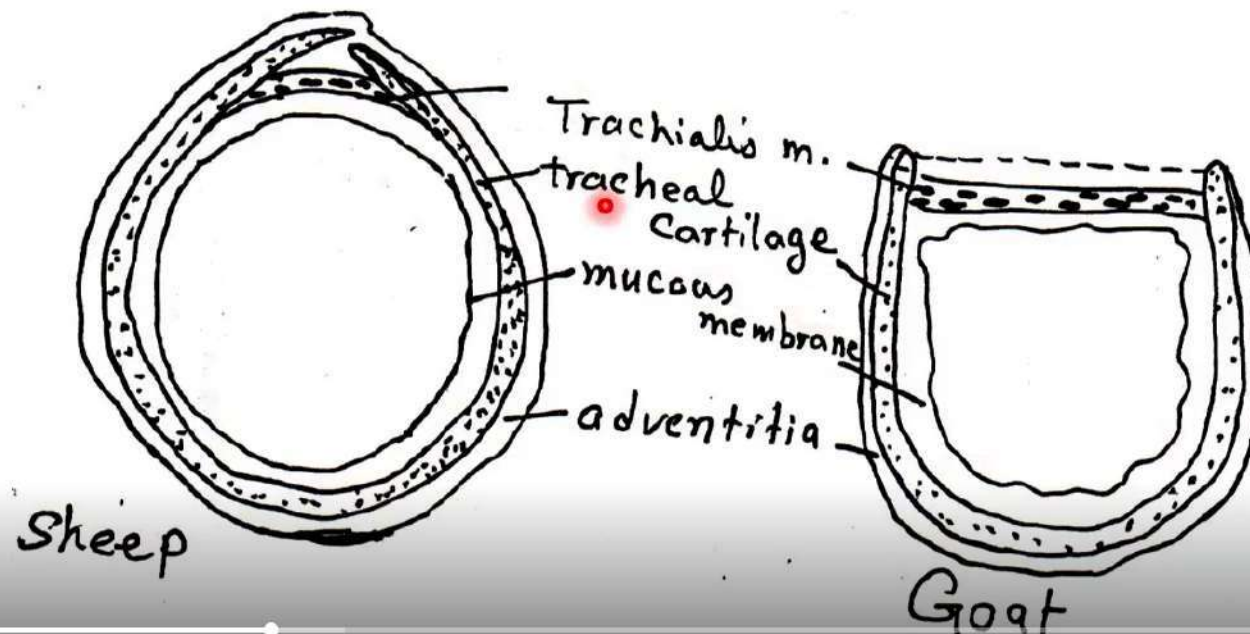


Sheep

- cylindrical
- 48-60 T-rings
- Cranial 1/3 of T-rings overlap but caudal 2/3 U-shape
- tracheal bronchus

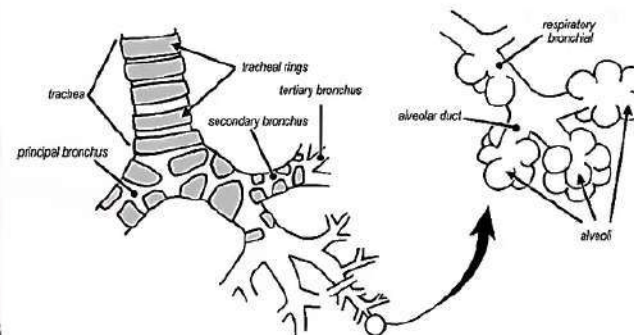
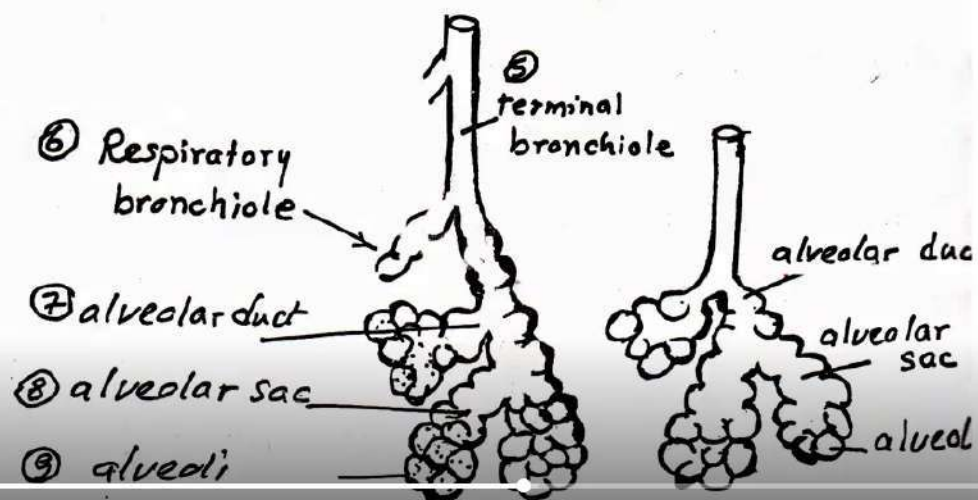
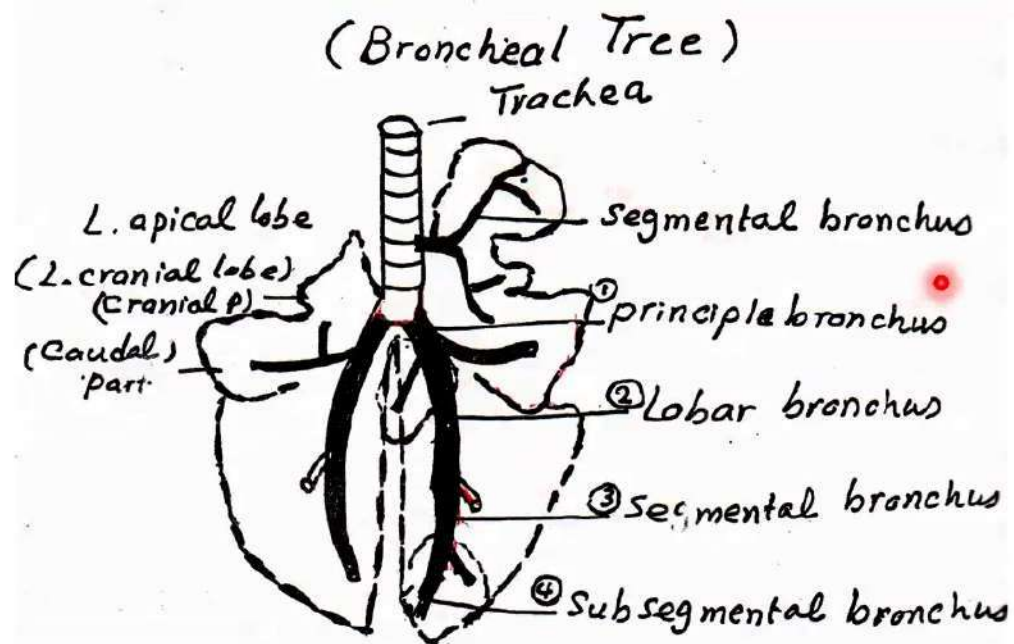
Goat

- U-shape
- 48-60 T-rings
- free ends of T-rings are widely separated by gap
- tracheal bronchus



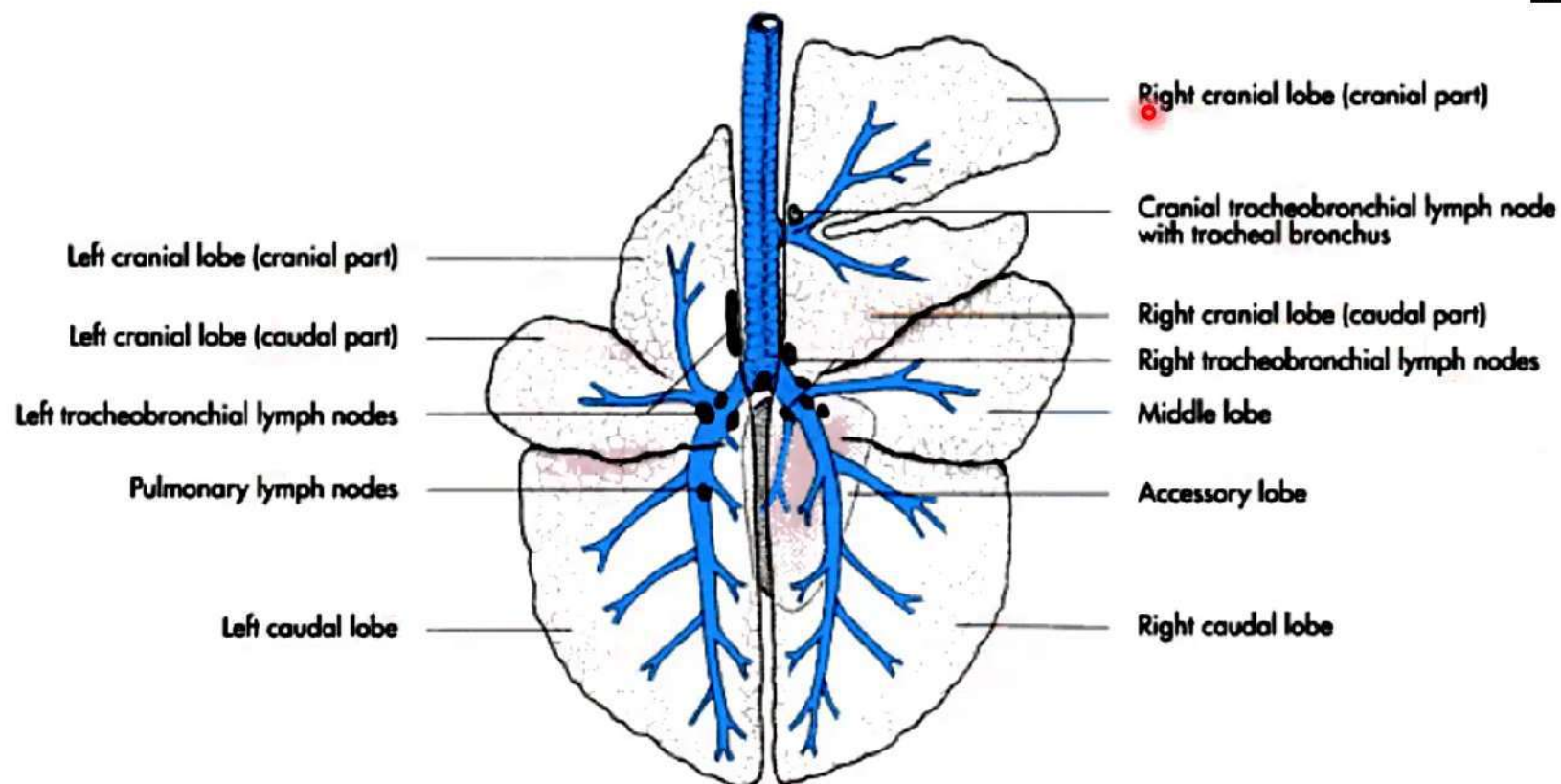


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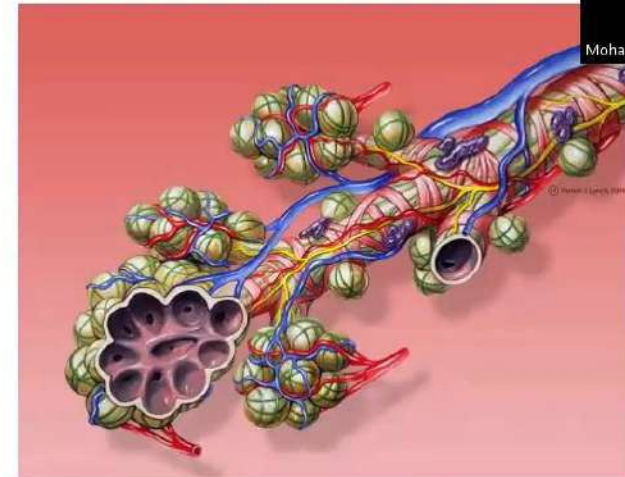
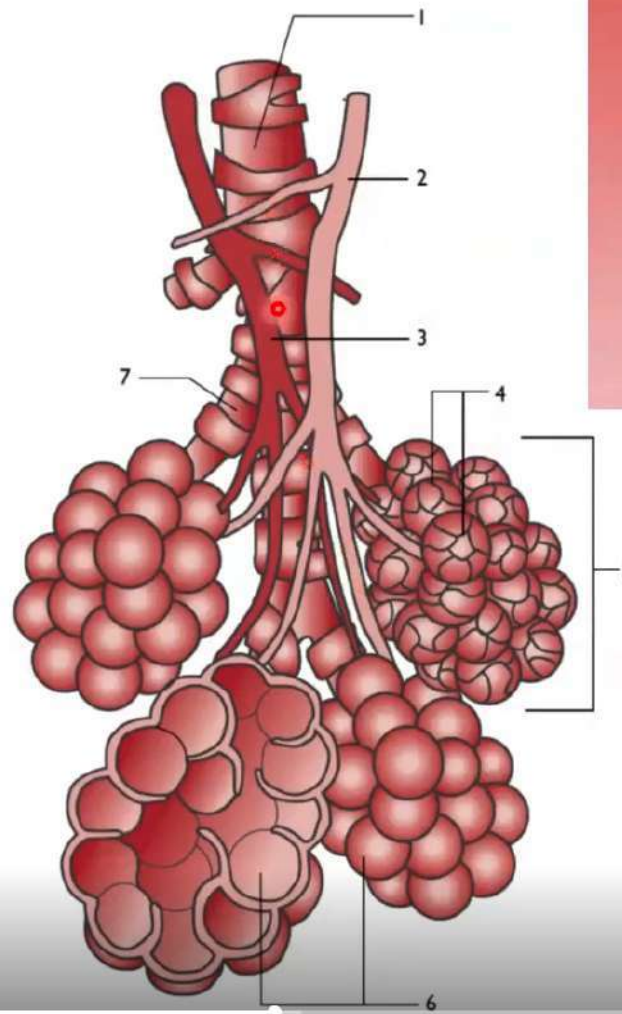




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- 1 – Terminal bronchiole
- 2 – Pulmonary arteriole
- 3 – Pulmonary venule
- 4 – Capillaries
- 5 – Alveolar sac
- 6 – Alveoli
- 7 – Alveolar duct



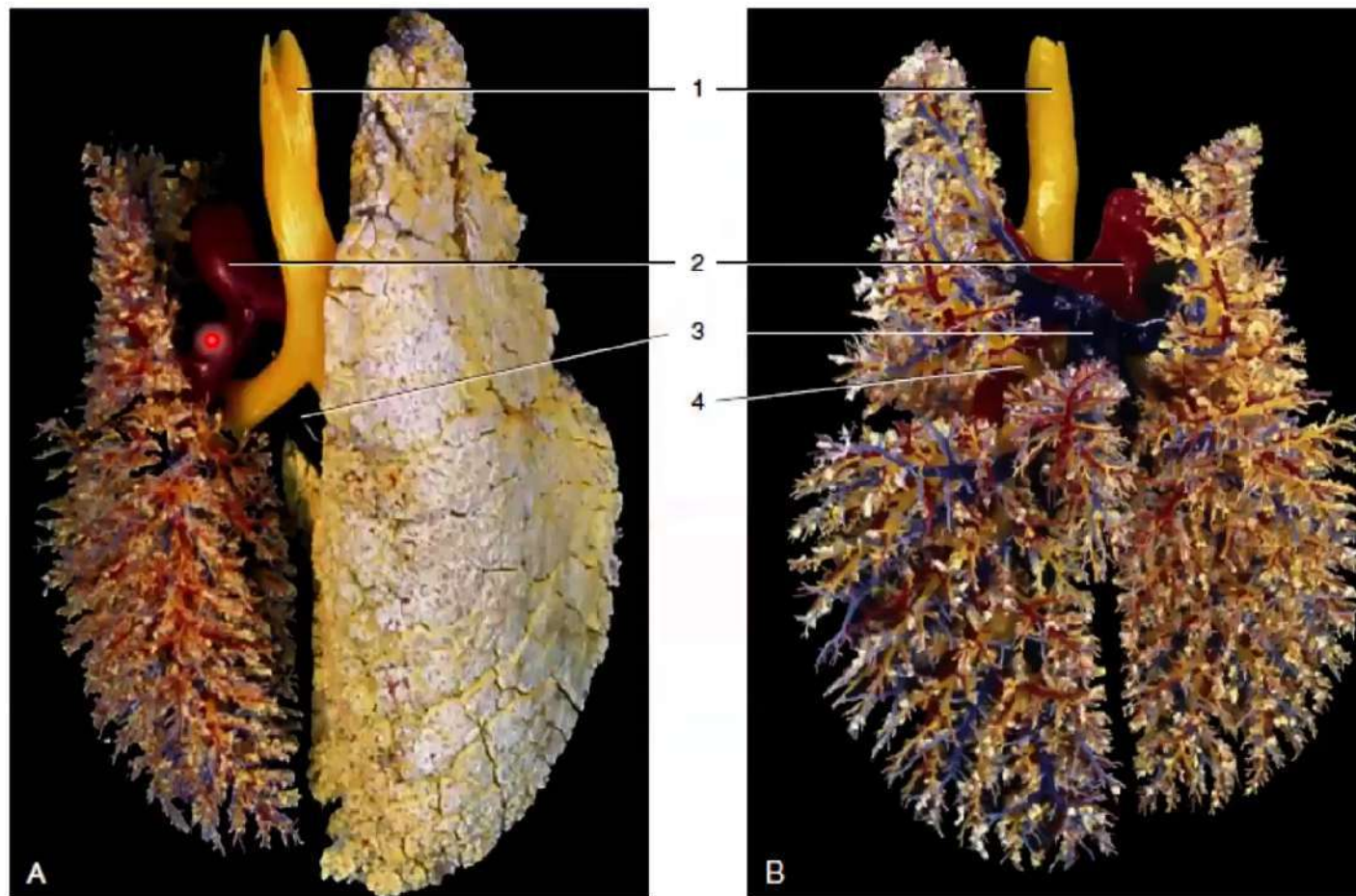
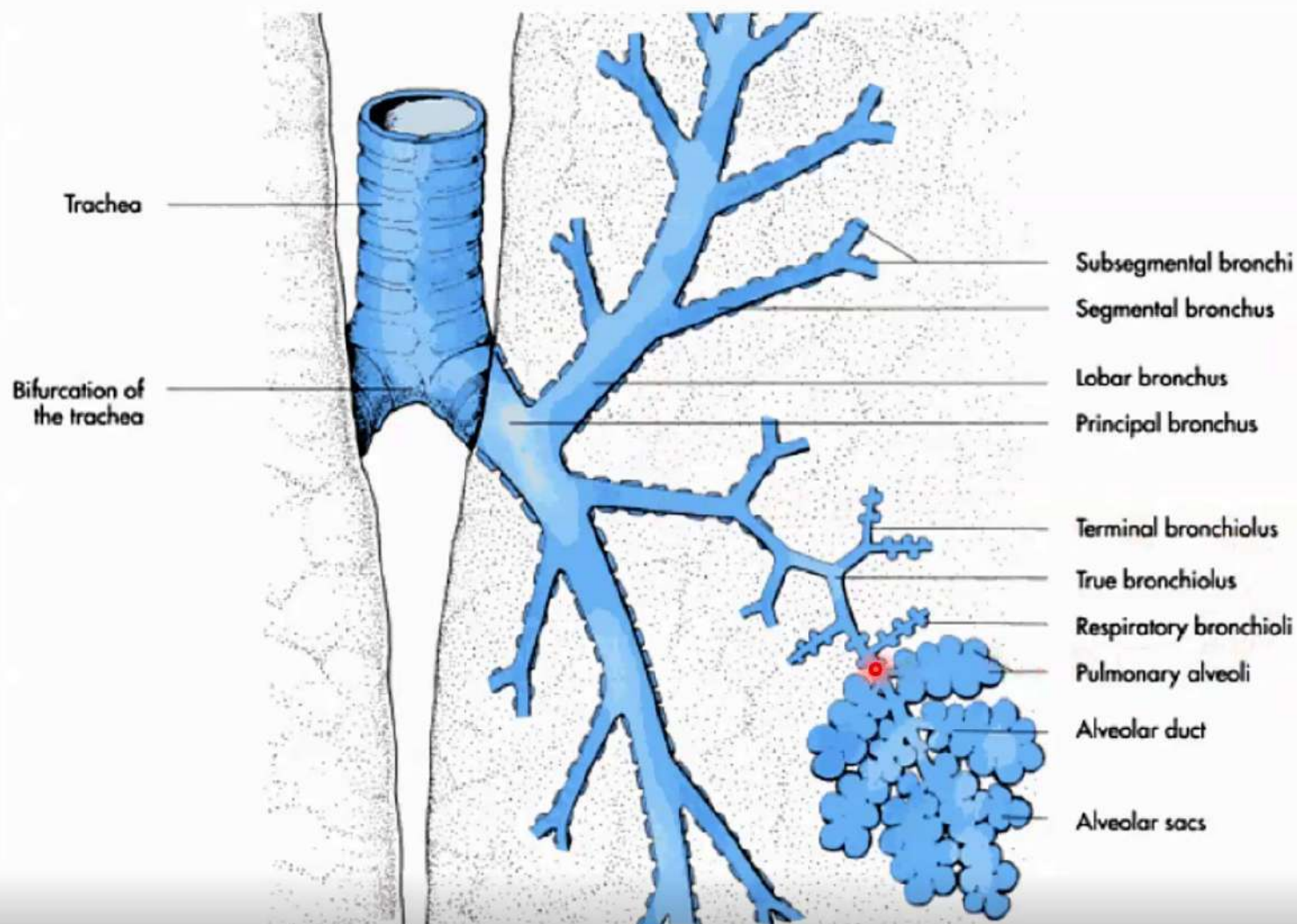


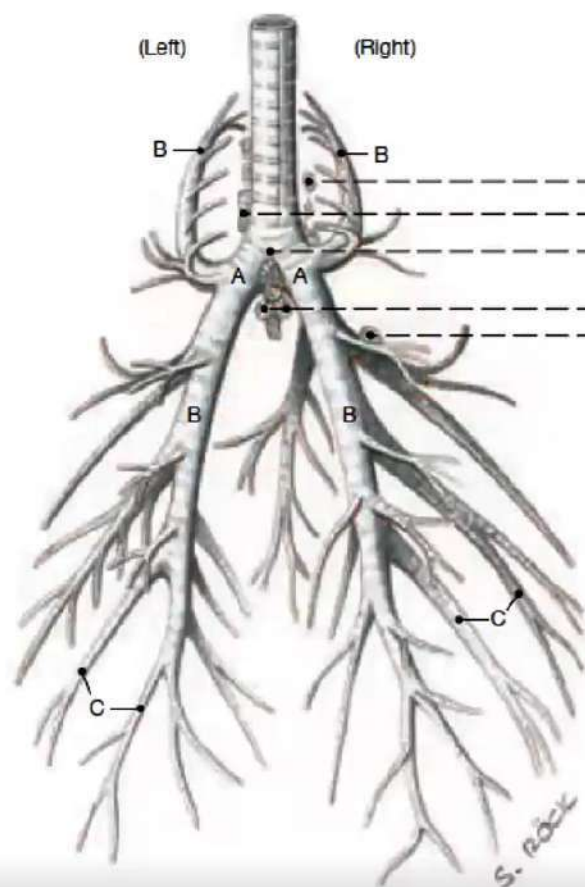
Figure 4-24 A, B, Dorsal view of the bronchial tree (*yellow*) and accompanying blood vessels of the pig (corrosion cast). 1, Trachea; 2, pulmonary trunk; 3, pulmonary veins; 4, tracheal bronchus.



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Bronchial Tree, dorsal view

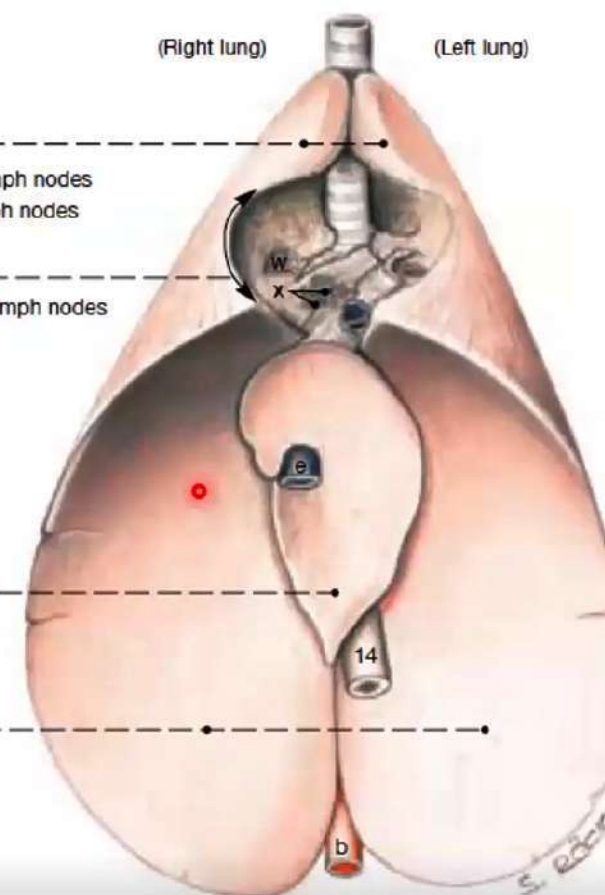


Lungs, ventral view

- 18 Cranial lobes
- 19 Right tracheobronchial lymph nodes
- 20 Left tracheobronchial lymph nodes
- 21 Tracheal bifurcation
- 22 Cardiac notch
- 23 Middle tracheobronchial lymph nodes
- 24 Pulmonary lymph nodes

- 25 Accessory lobe

- 26 Caudal lobes



Lung

- Shape
- Position
- 2 surface
- 3 border
- base
- apex
- Pulmonary hilus , pulmonary root , pulmonary ligament





Position:

- The right and left lungs occupy the greater part of the thoracic cavity.
- The lungs are located in the pleural sac.



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Fixation:

- It is attached by the root and pulmonary ligament with this exception, each lung lies free in the correspond pleural cavity.
- The root of the lung is invested by the pleura which extends caudally on each side of the triangular area of adhesion between the two lung, forming the pulmonary ligament.





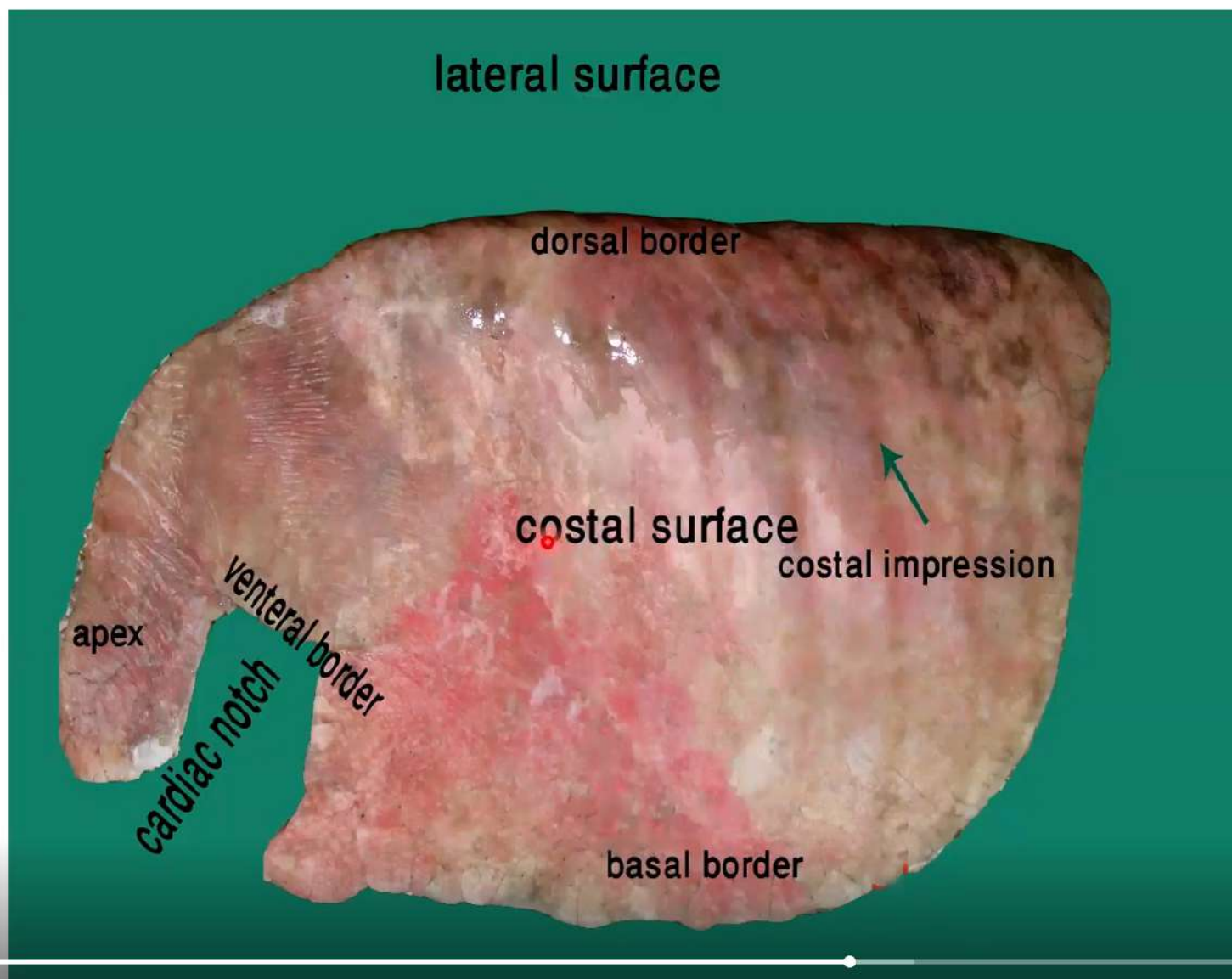
Shape:

- Each lung is semicone-shape.
- Each lung presents apex, base
 - Costal, mediastinal & diaphragmatic surface.
 - Dorsal, Ventral, & Medial border.

Colour:

It is varies according to amount of blood contained in them:

- During the life, the colour is pink.
- In an unbled animal is deep red.
- In an bled animal is almost white or faintly red.





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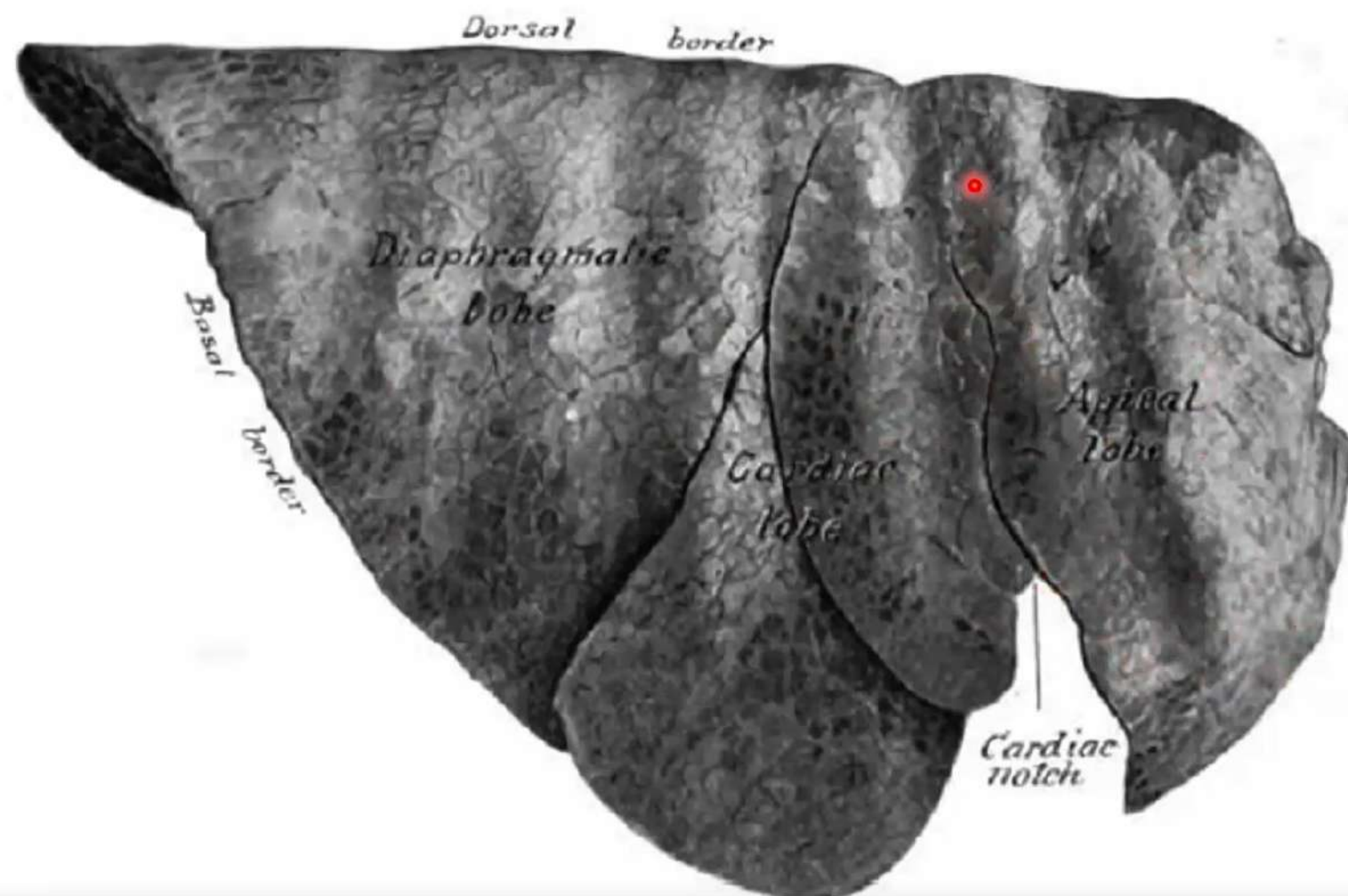
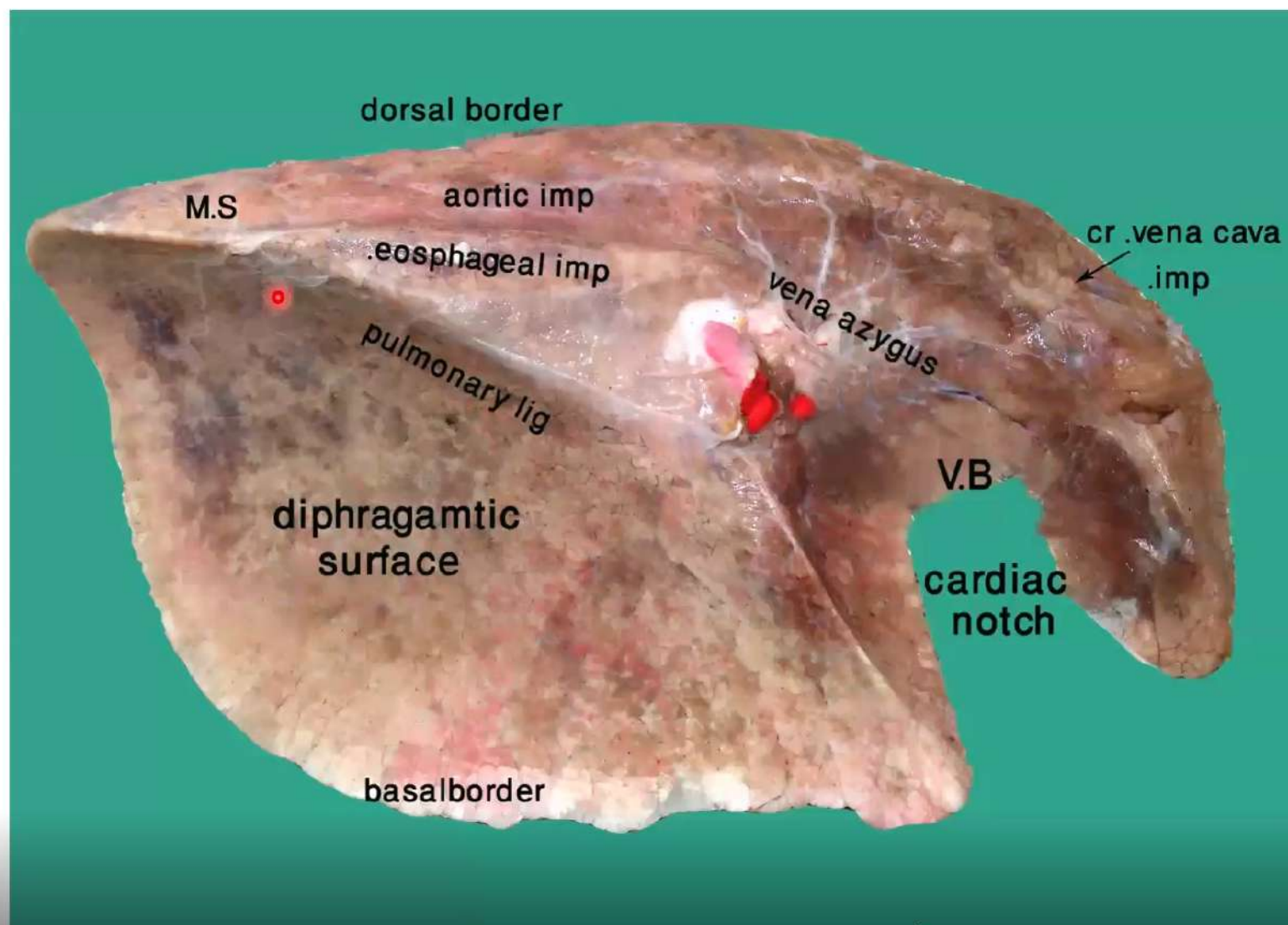


FIG. 367.—RIGHT LUNG OF OX, COSTAL SURFACE.
Organ hardened in situ.



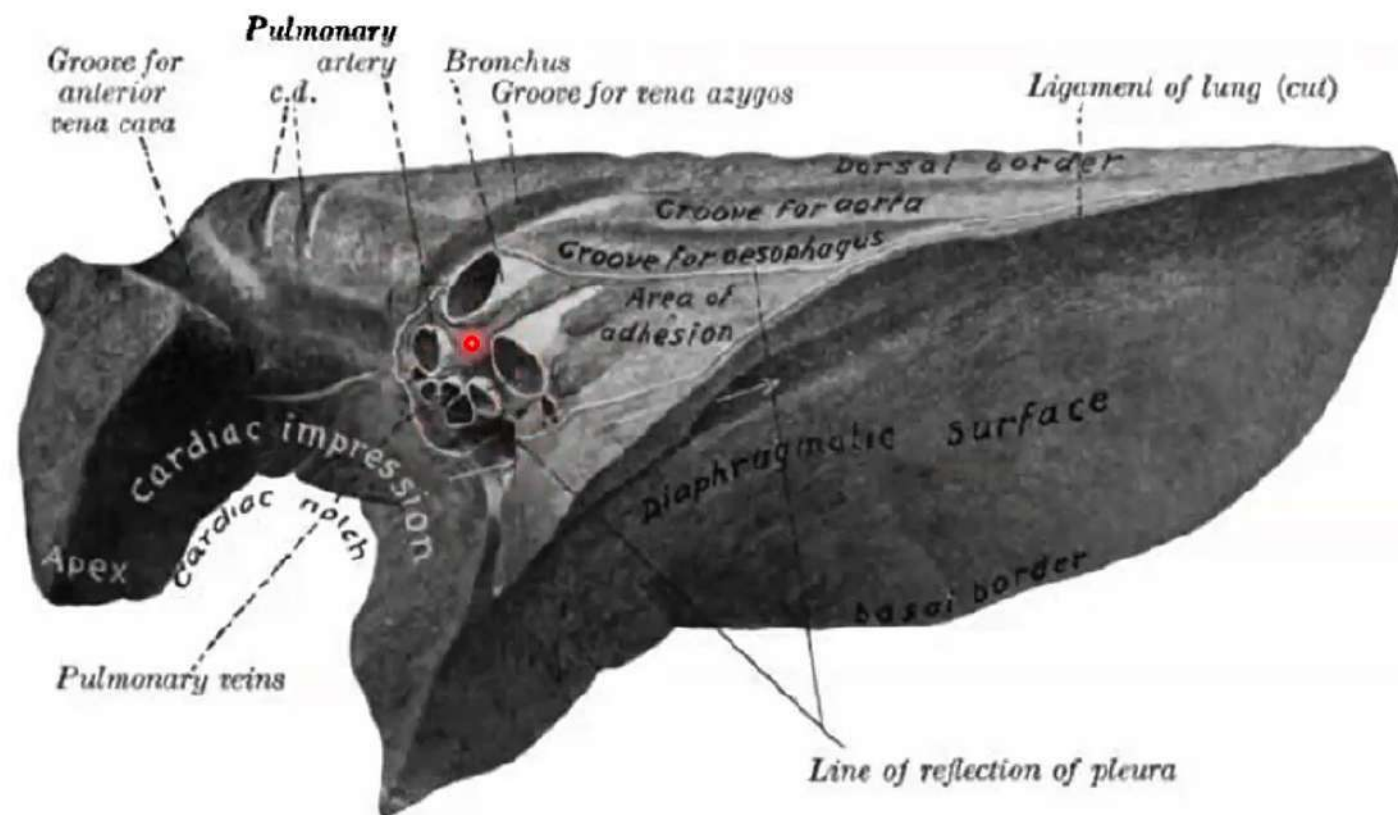


FIG. 363.—RIGHT LUNG OF HORSE, MEDIASTINAL AND DIAPHRAGMATIC SURFACES.

Organ hardened *in situ*. c, d, Grooves for superior cervical and dorsal veins. Arrows indicate canal for posterior vena cava between mediastinal lobe and main part of lung.



Pulmonary hilus: is the area at the middle of the mediastinal surface and behind the cardiac impression contain the root of the lung.

Root of lung (pulmonary root)

- all the structures entering or leaving the lung at the hilum invested with the pleura

- These structures comprise:**

- principal bronchus.
- Pulmonary artery (ventral to the bronchus).
- Pulmonary veins (ventral and caudal to P.A.).
- Bronchial artery (dorsal to the bronchus).
- Bronchial veins (dorsal to the bronchus).
- Pulmonary plexus of nerves.
- Lymph vessels and lymph node.

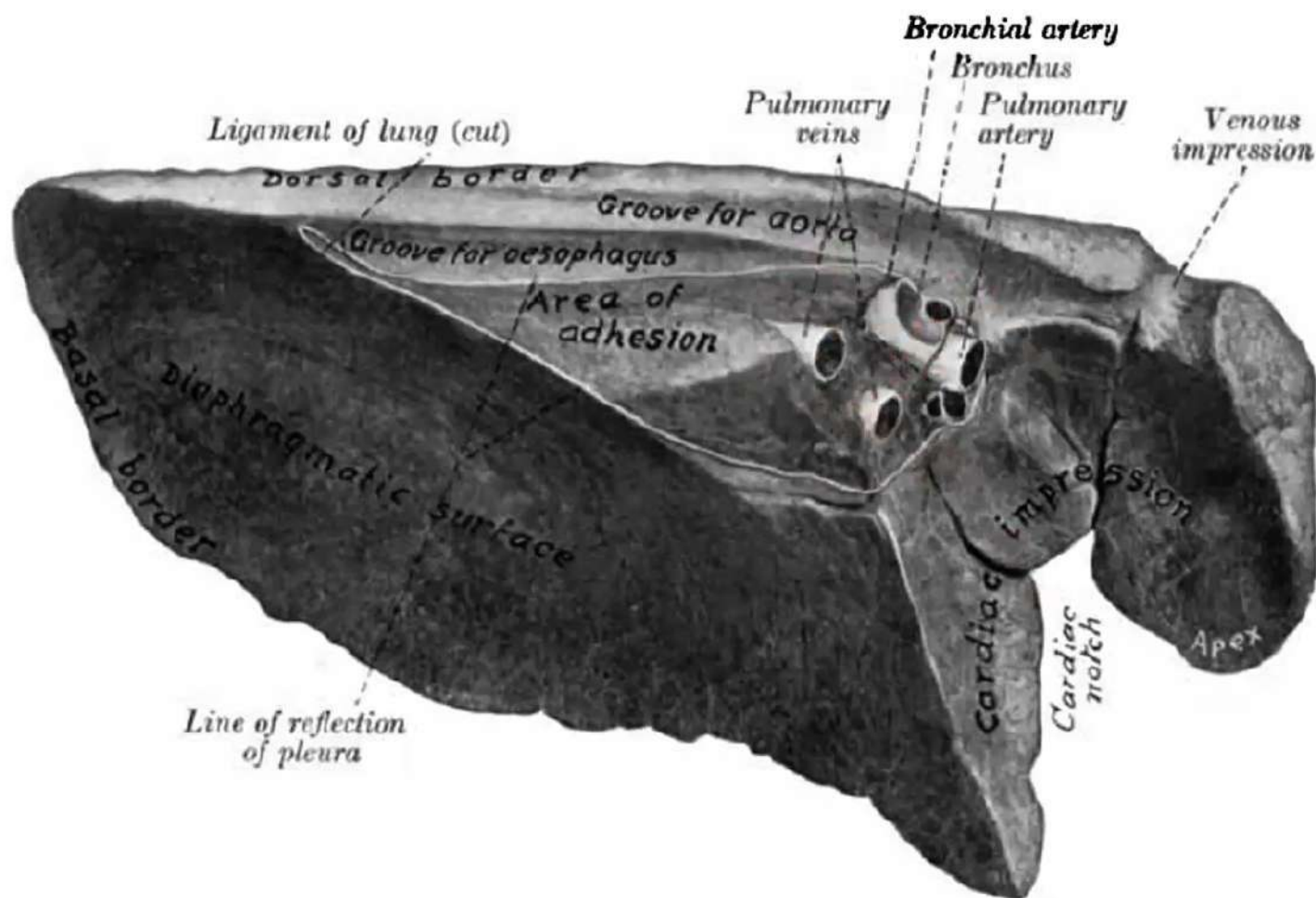


FIG. 362.—LEFT LUNG OF HORSE, MEDIASTINAL AND DIAPHRAGMATIC SURFACES.
Organ hardened *in situ*. (Venous impression for common dorso-cervico-vertebral stem.)

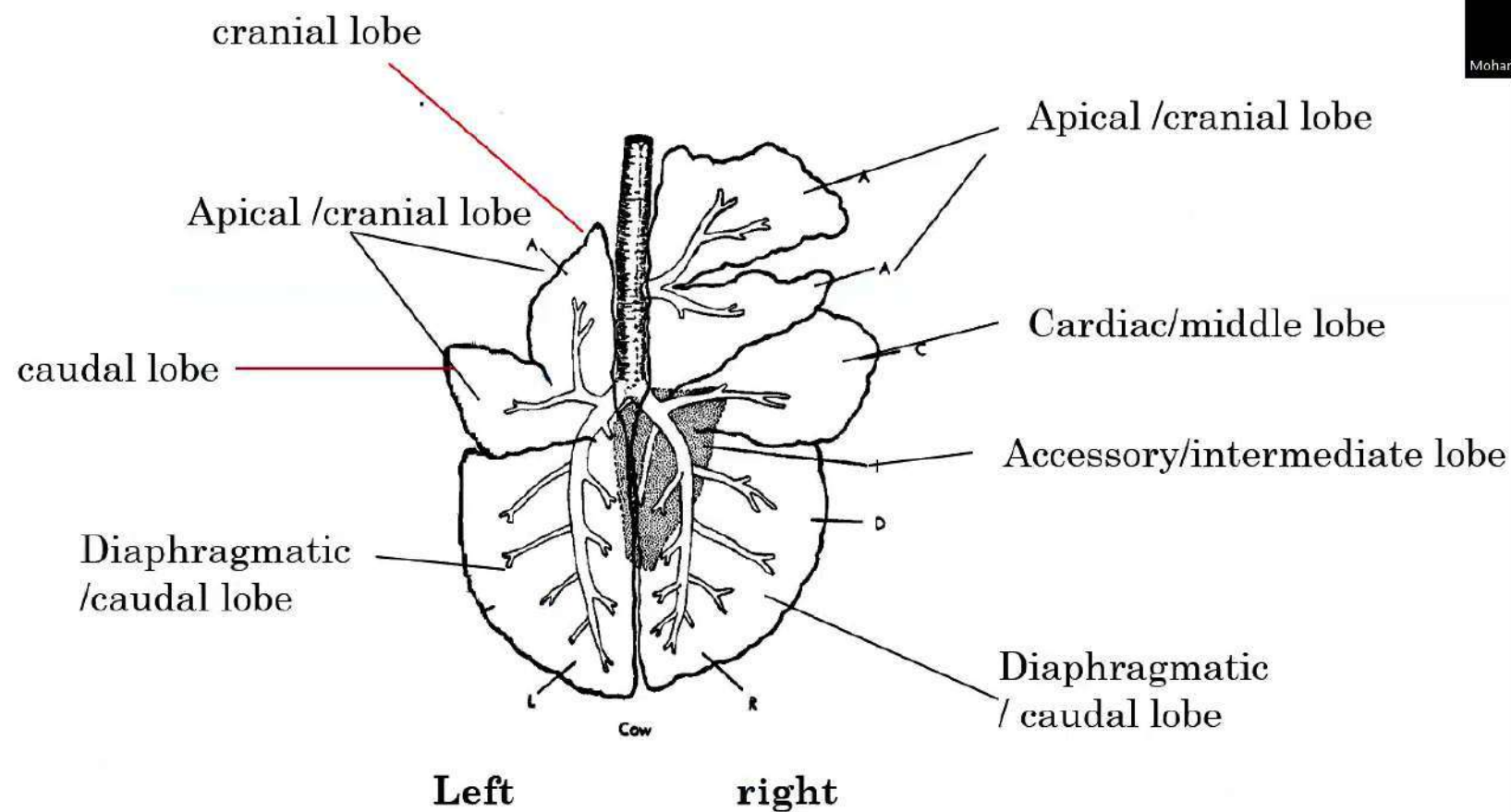
Lobes of the lung



- The lungs subdivided into lobes by fissures.
- ❖ In all animals except in horse, the right lung has 4 lobes:
 1. Apical (cranial).
 2. Middle (cardiac).
 3. Accessory.
 4. Diaphragmatic (caudal).
- ❖ In all animals the left lung has 2 lobes.
 1. Apical (cranial).
 2. Diaphragmatic (caudal).



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*Lobation of the lung of domestic Mammals .

Animal	Left lung	Right lung
Carnivores	Divided cranial lobe Caudal lobe	- Cranial lobe - Middle lobe - Caudal lobe - Accessory lobe
Camel	Lobes as horse	• tracheal bronchus • Pulmonary pleura fringes
Ruminants	As in dog	- Divided cranial lobe (Ventillated by tracheal bronchus) - Caudal lobe - Middle lobe - Accessory lobe
Horse	Cranial lobe Caudal lobe	- Cranial lobe - Caudal lobe - Accessory lobe .





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Item Animal	Left lung	Right lung	Interlobar fissure	Pulmonary fissure (lobulation)	Tracheal bronchus
Horse	Cranial lobe Caudal lobe	Cranial lobe Caudal lobe Accessory lobe	absent	Not distinct	absent
Ruminants	Cranial lobe divided Caudal lobe	Cranial lobe divided Middle lobe Caudal lobe Accessory lobe	Moderately deep, extend 2/3 to the dorsal border	Ox is very distinct Sheep is absent Goat very distinct	present
Camel	Cranial lobe Caudal lobe	Cranial lobe Caudal lobe Accessory lobe	absent	Very distinct	present
dog	Cranial divided Caudal lobe	Cranial lobe Middle lobe Caudal lobe Accessory lobe	Very deep reaching the principle bronchus	Not distinct	absent

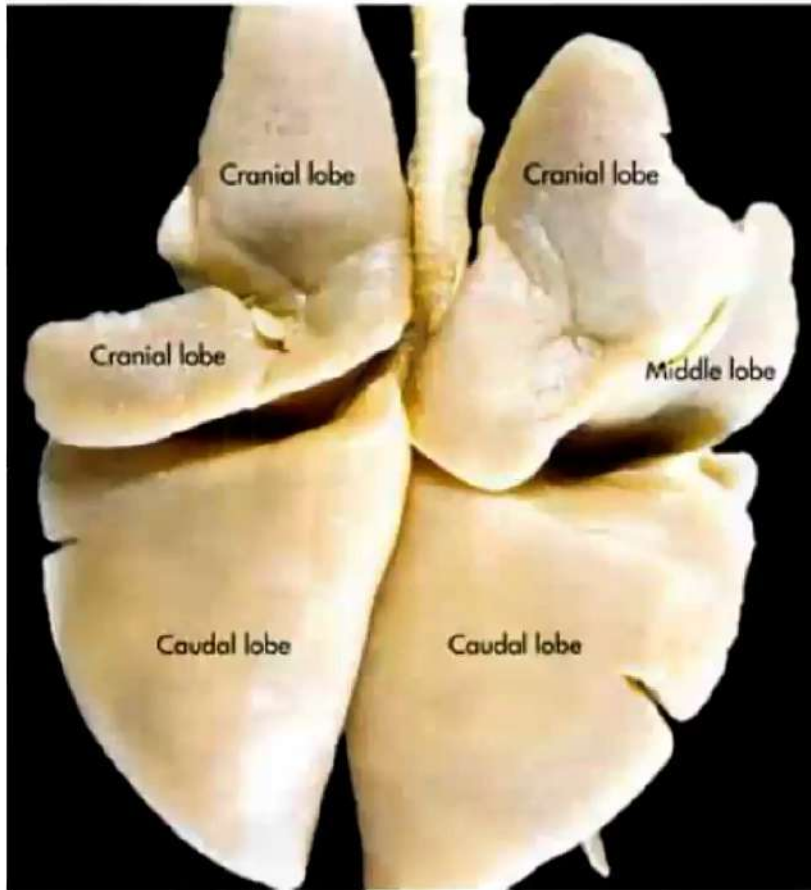
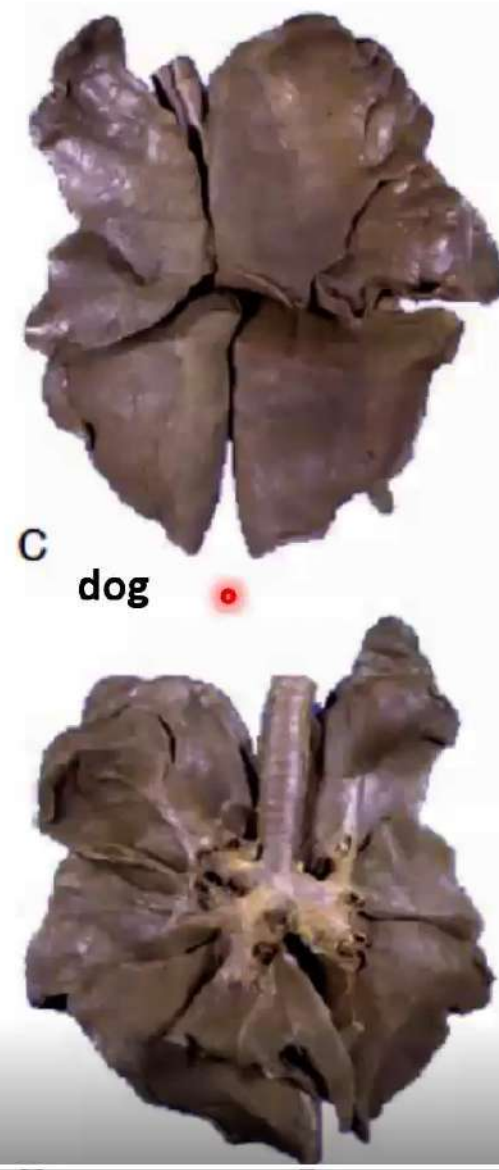
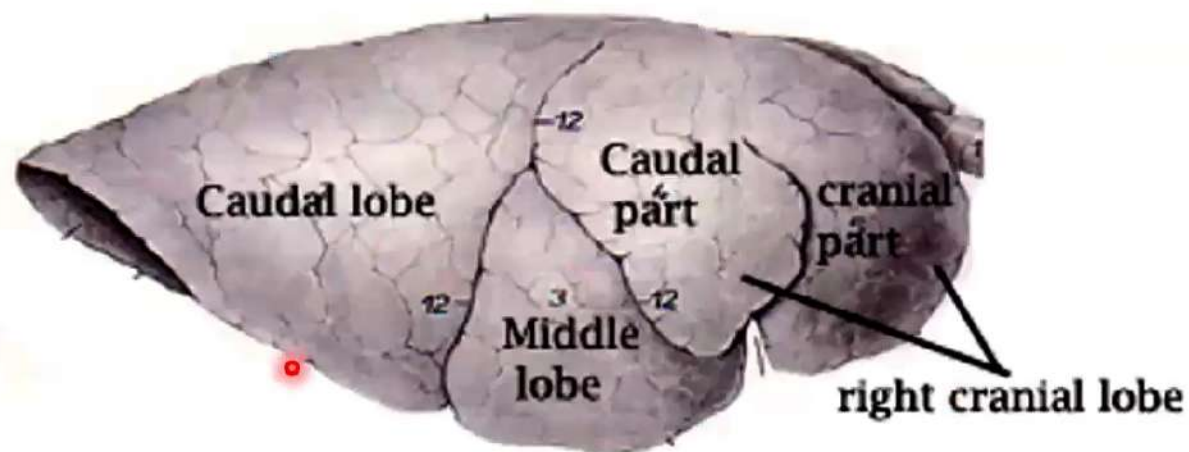


Fig. 8-33. Trachea and lung of a cat (dorsal aspect, S10 plastinate); courtesy of H. Dier, Vienna.

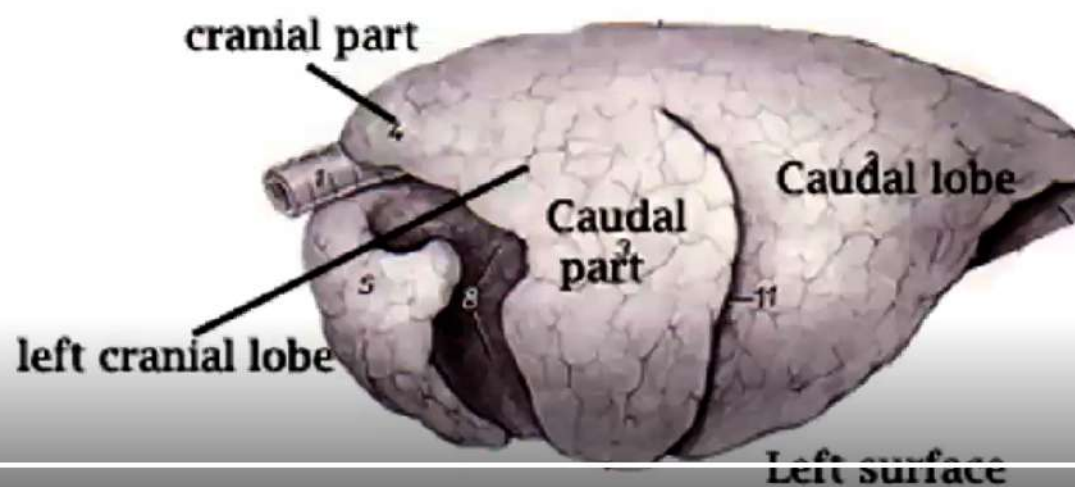


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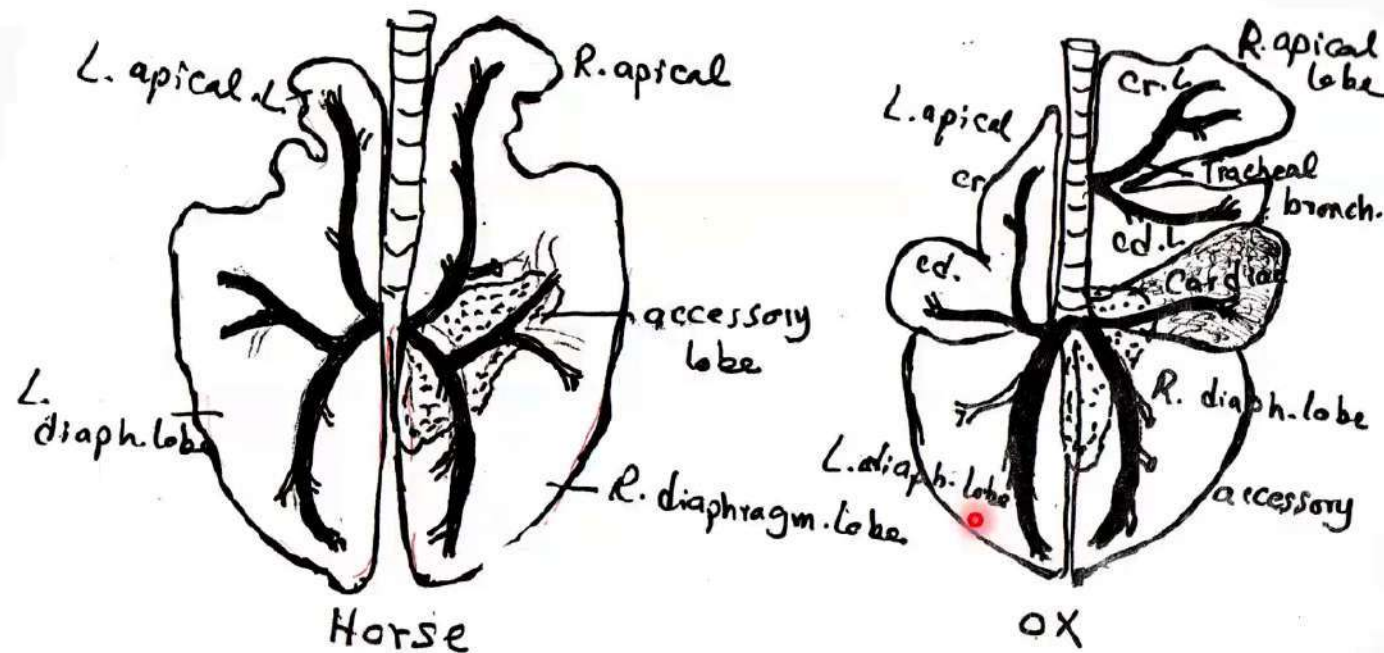
Lung of ox



Right surface



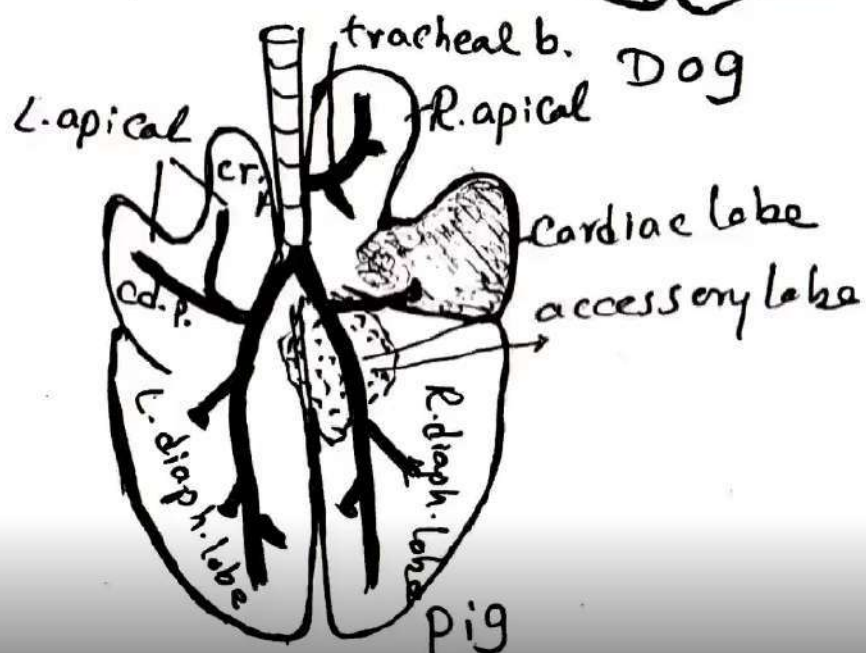
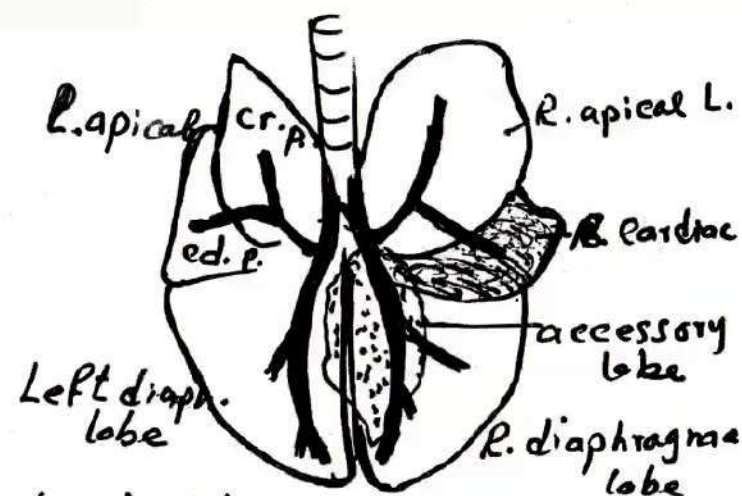
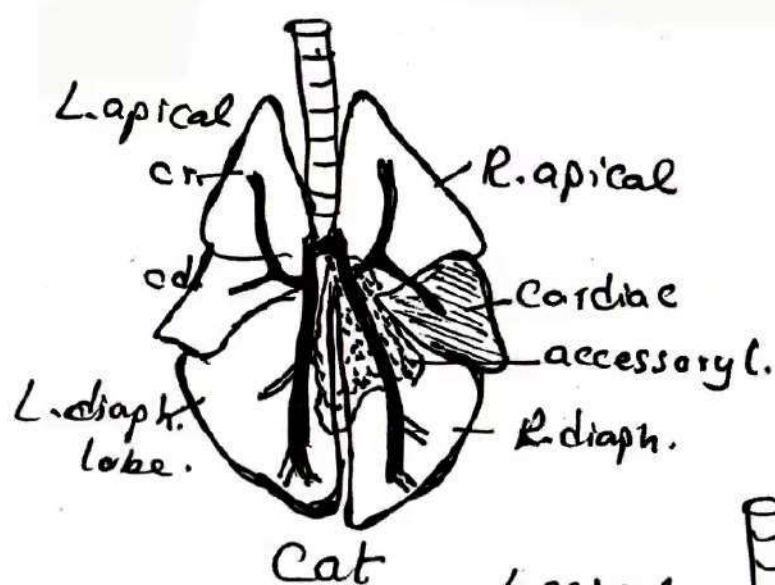
Left surface



Lobation of the lungs of domestic animals

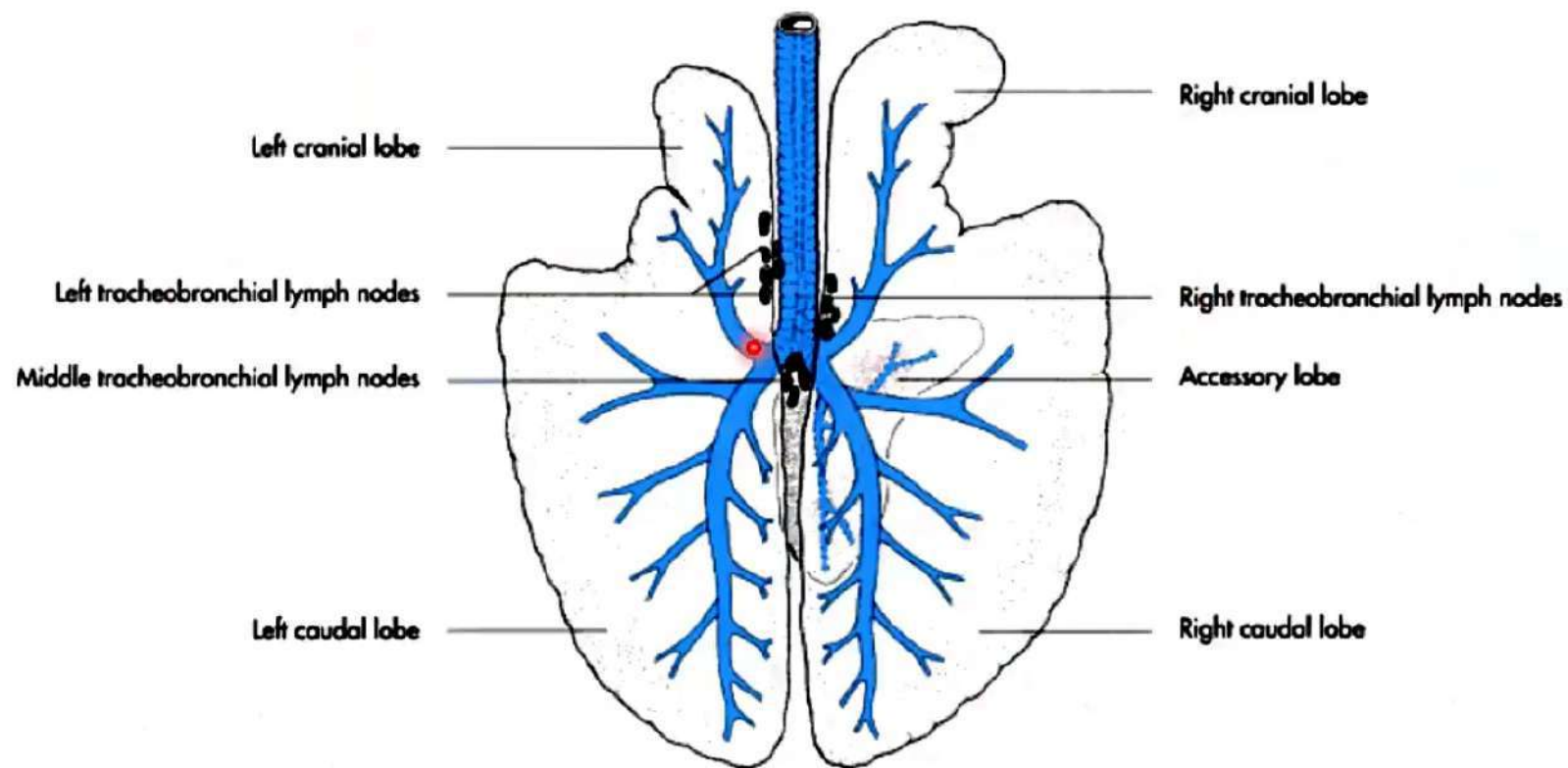


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Horse



The identification of the lungs of individual species is based on the degrees of lobation and lobulation.





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Horse	Ox	Camel
• No lobation • unclear lobulation	• clear lobation • clear lobulation • tracheal bronchus	• unclear lobation • clear lobulation • tracheal bronchus • Pulmonary pleura fringes

Small ruminants	Dog	Pig
• clear lobation • unclear lobulation • tracheal bronchus	• clear lobation • unclear lobulation	• clear lobation • moderate lobulation • tracheal bronchus

The blood supply of the lung may be divided into



Functional blood supply

It supply by venous blood through pulmonary arteries. It passes through the alveolar capillaries, and leaves lung through the pulmonary veins it nourishes the lung tissue distal to bronchioles.

Nutritional blood supply

It arise from broncho-esophageal and follows as bronchial arteries the ramifications of the bronchial tree, supplying the bronchi proximal to bronchioles.